



GE VERNOVA

ADMS

16.25.1

OMS Crew Modeling User Guide

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Change History

Date	Change Description
May 26, 2023	ADMS 3.16: • Updated Crew Modeling Summary Panel. • Updated Validating an Import File. • Editorial and formatting updates.
Oct 18, 2023	ADMS 16.1.0: • Updated Employees Tab. • Updated Crews Tab. • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Apr 1, 2024	ADMS 16.7.0: • Converted document to new format.
Sep 24, 2024	ADMS 16.12.0: • Updated section Crew Areas Tab.

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1. Crew Modeling Overview

Crew modeling is a part of the ADMS Outage Management System (OMS) package.

The OMS is designed to handle major storm situations by providing high-performance and effective tools combined with the ability to easily divide or combine areas of responsibility to balance the operator's workload. A comprehensive and flexible record-keeping function automatically provides all the necessary information for regulatory reporting requirements, reducing the amount of follow-up work after major storm events.

Assigning crews to outages and monitoring their current location and status can be performed directly on the outage management displays. Crew symbols can be positioned on the geographic network view or the outage information overview. Colors can be used to indicate crew status. Vehicle type, crew names, and assignment status are displayed as tool tips. A search facility allows for rapid identification of any crew by name, job, or truck ID.

The crew modeling tools and displays are used to build crews from scratch or remodel existing crews.

Crew Modeling tool is used to build a crew model. In addition, you can add employee and vehicle records ad hoc from the OMS Display plugin. For more information, refer to the *Outage Management System User Guide*.

Note

The Service Center ID fields must match in the crew and customer models, as they are used for **both** customers and crews. For more information about the customer model, refer to the *Customer List Compiler User Guide*.

1.1. General Philosophy

A crew model can be created by importing a tab delimited text file. Practically, a crew model is built by importing most of the data with a few manual entries for data that is not available from another system.

1.2. Crew Modeling Tool

The crew modeling application supports importing and exporting methods to directly open and save models. Changes in the crew model may be saved in the crew modeling tool by clicking the "Save" or "Save As" menu or toolbar item.

The static crew model is contained in a tab delimited "*.crew" text file with a journal format. Earlier versions of ADMS used a .txt extension for this file. The content is identical in both the *.txt and *.crew files.

The Crew Modeling tool supports importing and exporting the following information:

- Crew Model in XML format
- Employees
- Vehicles
- Crew Areas (optional)
- Employee Types (optional)
- Vehicle Types (optional)

Crews are created automatically based on fields in the employee records. There are no records for crews that can be imported.

Importing files for crew areas, employee types, and vehicle types is not required. Because the number of these records is small, it may be more efficient to type in the remaining fields once employees and vehicles are imported.

It is expected that the crew modeling tool is used to initially populate the crew model, and perhaps to periodically resynchronize with Human Resource records, or after a very large storm. Normally, the online crew displays are used for day-to-day maintenance, such as employees joining or leaving the company, transferring employees between crew areas, etc.

1.3. Crew Model and Import Philosophy

To create the initial crew model, the following steps are typically followed:

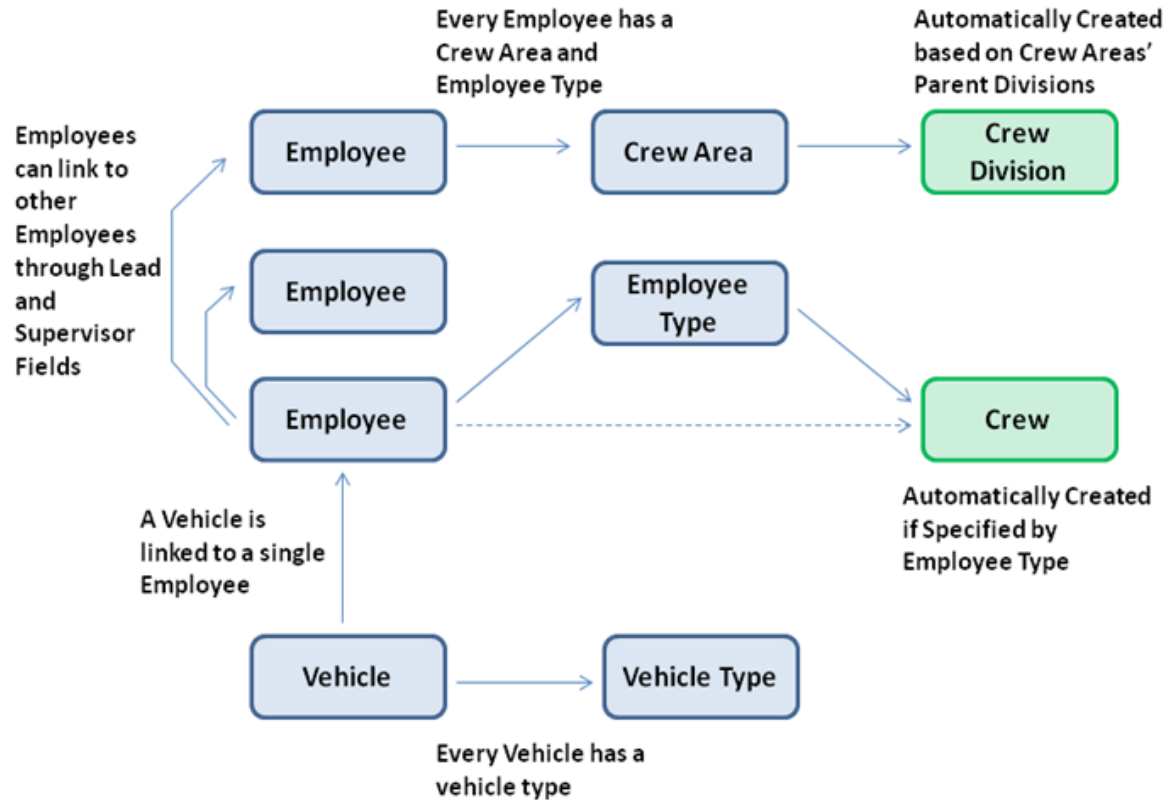
1. Import all employee input data, specifying at least the employee ID, the crew area, and the employee type. This step creates crew areas and employee types based on these employees.
2. Define fields on the crew area using the Crew Area tab.
3. Define fields on the employee type using the Employee Type tab.
4. Import or define additional employee fields that may come from a different source or have no source.
5. Import vehicles (optional). This step creates vehicle types. Vehicles are an optional feature.
6. Define fields on the vehicle type (optional).

The relationship between these records is shown in the figure below. Crews are displayed in the crew modeling tool automatically based on fields of the employee and employee type (if the employee type specifies to automatically create a crew).

When importing any file, the crew modeling tool adds the imported record if the ID (identifier) of the record does not exist; otherwise, the fields contained in the import file are updated. For example, it is possible to import employees by one crew division at a time, or to import radio numbers for employees by one crew division at a time.

You can create an import file by bringing the data into Excel and exporting this data to a .txt format with tab delimiters. The first row should contain column headers that match the field names that are expected on import (specified below), and subsequent rows contain data, one row per record.

The order of the columns is not significant, and extraneous columns are ignored when imported.



Crew Objects and Import

Figure 1. Crew Objects and Import

The modeling tool creates a default crew for every employee whose employee type specifies that a crew should be created. This employee is the "primary employee" of the crew.

An employee can be added to a default crew by setting the "lead" field of the employee to an employee that has an automated crew. The "supervisor" field of an employee causes sub-crews to be linked to the supervisor's crew.

A crew inherits attributes from the primary employee. For example, the crew area of a crew is set to the crew area of the primary employee. To change a crew's area, the crew area of the primary employee must be changed.

A crew contains at least one employee (the primary employee), but it can optionally have additional employees as members if their "lead" field is set. A crew inherits all vehicles that are linked to any employee on the crew. Temporary crews may be created in the online system only, and not in the modeling tool or display. However, the inheritance behavior of temporary crews is the same as default (or permanent) crews.

2. Crew Modeling Displays Overview

This section describes the fields and controls on the displays.

2.1. Crew Modeling Application Menu

The Crew Modeling application includes the following menu commands:

- **File > New:** Clears the fields to upload and build a new crew model.
- **File > Open:** Invokes the Open dialog box and prompts to upload a journal file (*.crew).
- **File > Export > Export XML:** Exports the currently open *.crew model file into an XML file.
- **File > Export > Export TSV:** Exports the currently open *.crew model file into multiple tab-separated value files.
- **File > Export > Export Log to TXT:** Exports log messages into a TXT file.
- **File > Import:** Contains the following import options:
 - **Import XML:** Imports all crew data from an XML file.
 - **Import Employees:** Imports employee data from a TXT file.
 - **Import Vehicles:** Imports vehicle data from a TXT file.
 - **Import Employee Types:** Imports employee types data from a TXT file.
 - **Import Vehicle Types:** Imports vehicle types data from a TXT file.
 - **Import Crew Areas:** Imports crew areas data from a TXT file.
- **File > Save:** Saves the changes into the current .crew file.
- **File > Save As:** Saves the .crew file with a new name.
- **File > Exit:** Exits the application.

2.2. Crew Modeling Summary Panel

Use the Summary panel to filter records on the Crew Modeling tabs by crew divisions.

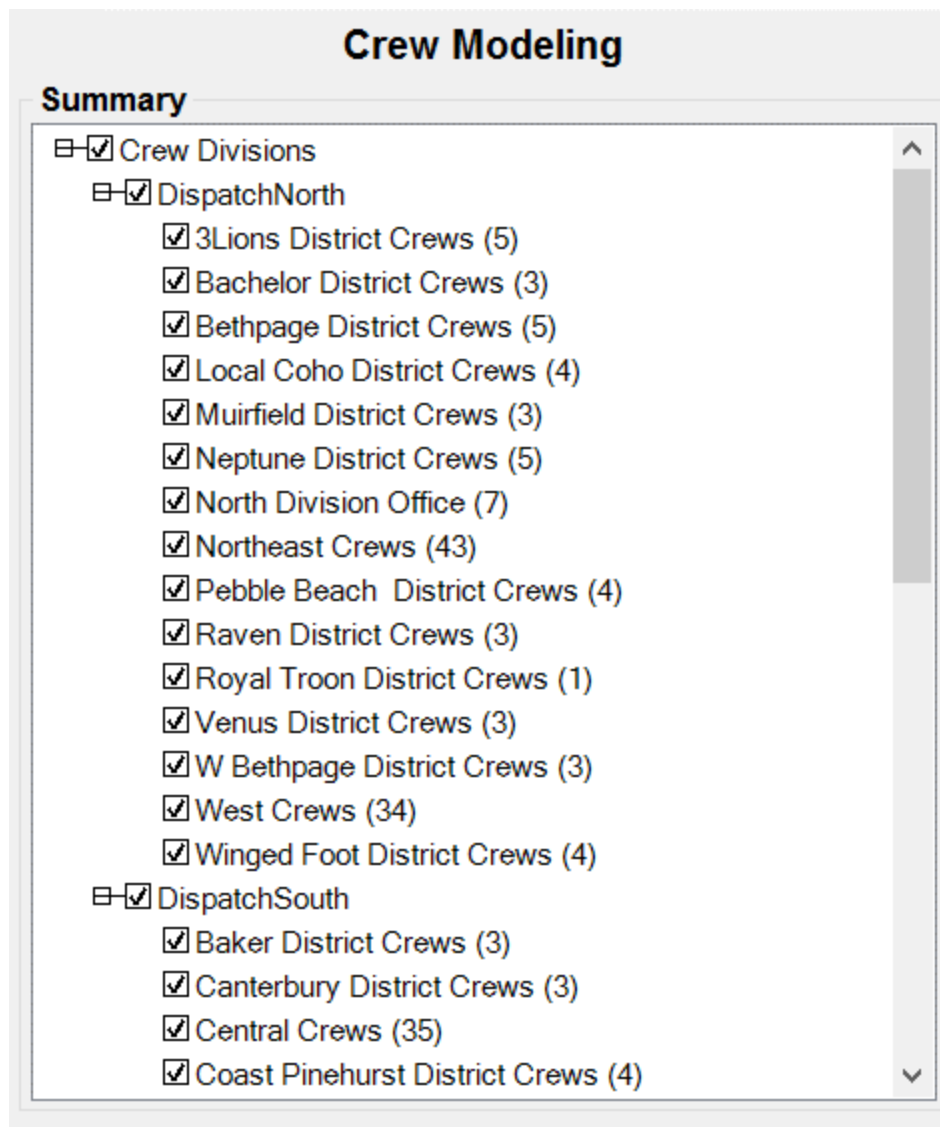


Figure 2. Crew Modeling Summary Panel

Select a crew division box to include the crew data in the tabbed view.

2.3. Employees Tab

Use the Employees tab to view and update the employee information.

Employees Crews Employee Types Crew Areas Vehicle Types Vehicles Messages

Crew Areas: Employee Types:

Employee Status

☐									
☐	Id	FirstName	Last Name	Members Count	Radio Number	Crew Division	Crew Area	Default Crew Area	Employee Type
☐	FI1571	Ethan G	Bertilson	0	1100	DispatchNorth	3Lions Distri...	3Lions Distri...	Local Lineman
☐	FI2576	Mary I	Crosby	0		DispatchNorth	3Lions Distri...	3Lions Distri...	Supervisor
☐	FI1502	Daniel Jeffrey	Dodge	0	1101	DispatchNorth	3Lions Distri...	3Lions Distri...	Local Lineman
☐	FI2625	Ryan A	Gorman	0		DispatchNorth	3Lions Distri...	3Lions Distri...	Administrator
☐	FI2685	Bonnie L	Persons	0		DispatchNorth	3Lions Distri...	3Lions Distri...	Manager
☐	FI1513	Sam A	Sheridan	0	1102	DispatchNorth	3Lions Distri...	3Lions Distri...	Local Lineman
☐	FI1564	Ron X	Stanton	0	1103	DispatchNorth	3Lions Distri...	3Lions Distri...	Local Lineman
☐	FI1520	William T	Webster	0	1104	DispatchNorth	3Lions Distri...	3Lions Distri...	Local Lineman
☐	FI1515	Eugene B	Graziano	0	1122	DispatchNorth	Bachelor Dis...	Bachelor Dis...	Local Lineman
☐	FI1578	Sean H	Jenson	0	1123	DispatchNorth	Bachelor Dis...	Bachelor Dis...	Local Lineman
☐	FI2692	Brandon Mich...	Kreider	0		DispatchNorth	Bachelor Dis...	Bachelor Dis...	Manager
☐	FI2638	G Patrick	Montstream	0		DispatchNorth	Bachelor Dis...	Bachelor Dis...	Administrator
☐	FI1527	Brent L	Myers	0	1124	DispatchNorth	Bachelor Dis...	Bachelor Dis...	Local Lineman
☐	FI1511	Rusty Jim	Brewer	0	1125	DispatchNorth	Bethpage Di...	Bethpage Di...	Local Lineman
☐	FI1528	Bret V	Cooper	0	1126	DispatchNorth	Bethpage Di...	Bethpage Di...	Local Lineman
☐	FI1584	Chris B	Hallett	0	1127	DispatchNorth	Bethpage Di...	Bethpage Di...	Local Lineman
☐	FI1510	L Brian	Karlak	0	1128	DispatchNorth	Bethpage Di...	Bethpage Di...	Local Lineman
☐	FI2693	Diane Eykholt	Lelivelt	0		DispatchNorth	Bethpage Di...	Bethpage Di...	Manager
☐	FI2644	Michaeline O	Myers	0		DispatchNorth	Bethpage Di...	Bethpage Di...	Administrator

Figure 3. Employees Tab

The Employee Status display shows the following information:

- **Select:** Select this check box to select the employee record.
- **ID:** Employee's identification number.
- **First Name:** Employee's first name.
- **Last Name:** Employee's last name.
- **Members Count:** Number of members that are associated with this employee. This option is used for foreign (mutual aid) employees.
- **Radio Number:** Employee's radio number.
- **Crew Division:** The division the crew operates within.
- **Current Crew Area:** Current crew area.
- **Default Crew Area:** Default crew area.
- **Employee Type:** The employee's classification (for example, Lineman or Apprentice).
- **Lead:** Name of the employee's lead.
- **Supervisor:** Name of the employee's supervisor.
- **Is Evaluator:** Flag to indicate if the employee is an evaluator.
- **Department:** Name of the department the employee belongs to.
- **Can Switch:** A flag to indicate that the employee is allowed to operate a switch.
- **Auto Dispatch ID:** The unique identification number from the Work Force Management system to track the crew's coordinates and communicate the crew location to the OMS system.

- **Is Primary:** Indicates whether this employee is assigned as a primary employee for a particular crew.
- **MDT Flag:** Indicates whether the vehicle to which this employee is assigned is equipped with a mobile data terminal.

Important

To enable integration with the Crew Server web interface and Workforce Management System, the MDT Flag field must be set to True and the Auto Dispatch ID field must be populated for primary employees of crews that are equipped with a Mobile Data Terminal (MDT).

- **Status:** Employee's status.
- **Phone Number:** Employee's phone number.
- **Secondary Phone Number:** Employee's secondary phone number.
- **Capability:** Describes the employee's capability (switching, can work on incident).
- **Temp Area:** Area the crew is temporarily assigned to.
- **Auto Dispatch:** Indicates if the crew is enabled for the auto dispatch process.

2.3.1. Employee Tab Context Menu

You can access this context menu by right-clicking on any employee record.

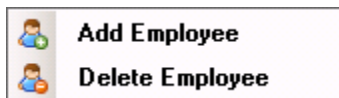


Figure 4. Employee Tab Context Menu

The following options are available:

- **Add Employee:** Activates the Add Employee pane, where you can create a record for a new employee.
- **Delete Employee:** Deletes the selected employee record.

When several records are selected, the context menu options include:

- **Delete Employees:** Deletes all selected employees.

2.3.2. Add/Update Employee Pane

Use the Add Employee pane to create a new employee record. Use the Update Employee pane to change the selected employee's information.

- To display the Add Employee pane, select the Add Employee option on the Employee tab's context menu.

The required fields to add an employee are: Crew Division, Current and Default Crew Area, Employee Type, First Name, Last Name, and ID.

- To display the Update Employee pane, click on an employee record on the Employee tab.

Update Employee

Properties

Name	Value
Automatic Dispatch Id	11100
Capability	CanWorkOnIncident ▼
Crew Area Id	LL1000
Crew Division	DispatchNorth ▼
Crew Name	Ethan G Bertilson
Current Crew Area	3Lions District Crews ▼
Default Crew Area	3Lions District Crews ▼
Default Lead Name	<None> ▼
Department	Dist Ops
Employee First Name	Ethan G
Employee Last Name	Bertilson
Employee Type	Local Lineman ▼
Employee Type Id	3100
Id	FI1571
Is Available	True ▼
Is Contractor	False ▼

Figure 5. Update Employee Pane

The Update Employee pane contains the following fields:

- **Automatic Dispatch ID:** The automatic dispatch ID of an employee.
- **Capability:** Describes the employee's capability (switching, can work on incident).
- **Crew Area ID:** If the employee is assigned to a crew, the headquarters ID to which the crew belongs.
- **Crew Division:** The division the crew belongs to.
- **Crew Name:** Name of the crew that the employee belongs to. The crew name is populated with the name of the employee's default lead.
- **Current Crew Area:** Current crew area polygon.
- **Default Crew Area:** Default crew area polygon.
- **Default Lead Name:** Name of the employee's default lead.
- **Department:** Name of the department the employee belongs to.

- **Employee First Name:** The employee's first name.
- **Employee Last Name:** The employee's last name.
- **Employee Type:** The employee's classification (for example, Lineman or Apprentice).
- **Employee Type ID:** The employee's classification ID.
- **ID:** The employee's identification number.
- **Is Available:** Indicates whether this employee is available for an assignment.
- **Is Contractor:** Indicates whether this employee is a contractor.
- **Is Evaluator:** Indicates whether this employee is an evaluator.
- **Is Foreign:** Indicates whether this employee reports to another company. In a storm situation, foreign crews may be required to restore the power.
- **Is Primary:** Indicates whether this employee is assigned as a primary employee for a particular crew.
- **Mobile Dispatch Enabled:** Indicates whether the vehicle to which this employee is assigned is equipped with a mobile data terminal (MDT).

Important

To enable integration with the Crew Server web interface and Workforce Management System, the Mobile Dispatch Enabled field must be set to True and the Automatic Dispatch ID field must be populated for primary employees of crews that are equipped with a mobile data terminal (MDT).

- **Phone Number:** The employee's primary phone number.
- **Radio Number:** The radio number assigned to the employee.
- **Secondary Phone Number:** The employee's alternate phone number.
- **Status:** Indicates if the employee is available for duty (active, inactive, on call).
- **Supervisor Name:** Name of the employee's supervisor.
- **Team Members Count:** Number of members that are associated with this employee. This option is used for foreign (mutual aid) employees.
- **Temp Crew Name:** Name of the temporary crew the employee belongs to. The name is populated with the last name and ID of the temporary crew primary employee (for example, Walker [TEMPFI2791]).
- **Vehicle IDs:** Identification numbers of the default vehicles to which the employee is assigned.
- **Cancel:** Click to discard the changes and close the Add/Update pane.
- **OK:** Click to save the changes.

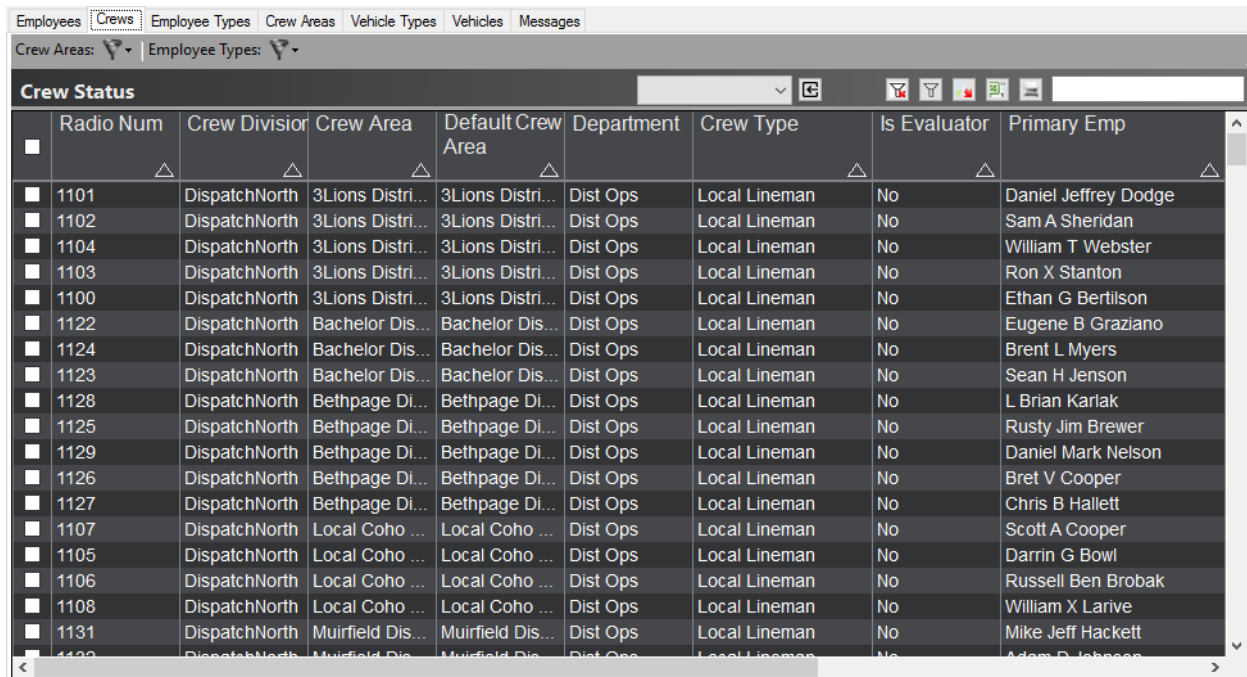
The **Add Employee** pane has the same fields as the **Update Employee** pane and also includes the following fields:

- **Department ID:** Identification number of the department the employee belongs to.

- **Current Lead Name:** Name of the employee's current lead.
- **Default Crew Area ID:** Identification number of default crew polygon.
- **Supervisor ID:** Identification number of the employee's supervisor.
- **Vehicles:** Vehicle types.

2.4. Crews Tab

Using the Crew Status display, you can update and manage existing crews.



The screenshot shows the 'Crews' tab selected in a software interface. Below the tab are filters for 'Crew Areas' and 'Employee Types'. The main area is titled 'Crew Status' and contains a table with the following columns: Radio Num, Crew Division, Crew Area, Default Crew Area, Department, Crew Type, Is Evaluator, and Primary Emp. The table lists 17 crew records with their respective details.

	Radio Num	Crew Division	Crew Area	Default Crew Area	Department	Crew Type	Is Evaluator	Primary Emp
☐	1101	DispatchNorth	3Lions Distri...	3Lions Distri...	Dist Ops	Local Lineman	No	Daniel Jeffrey Dodge
☐	1102	DispatchNorth	3Lions Distri...	3Lions Distri...	Dist Ops	Local Lineman	No	Sam A Sheridan
☐	1104	DispatchNorth	3Lions Distri...	3Lions Distri...	Dist Ops	Local Lineman	No	William T Webster
☐	1103	DispatchNorth	3Lions Distri...	3Lions Distri...	Dist Ops	Local Lineman	No	Ron X Stanton
☐	1100	DispatchNorth	3Lions Distri...	3Lions Distri...	Dist Ops	Local Lineman	No	Ethan G Bertilson
☐	1122	DispatchNorth	Bachelor Dis...	Bachelor Dis...	Dist Ops	Local Lineman	No	Eugene B Graziano
☐	1124	DispatchNorth	Bachelor Dis...	Bachelor Dis...	Dist Ops	Local Lineman	No	Brent L Myers
☐	1123	DispatchNorth	Bachelor Dis...	Bachelor Dis...	Dist Ops	Local Lineman	No	Sean H Jenson
☐	1128	DispatchNorth	Bethpage Di...	Bethpage Di...	Dist Ops	Local Lineman	No	L Brian Karlak
☐	1125	DispatchNorth	Bethpage Di...	Bethpage Di...	Dist Ops	Local Lineman	No	Rusty Jim Brewer
☐	1129	DispatchNorth	Bethpage Di...	Bethpage Di...	Dist Ops	Local Lineman	No	Daniel Mark Nelson
☐	1126	DispatchNorth	Bethpage Di...	Bethpage Di...	Dist Ops	Local Lineman	No	Bret V Cooper
☐	1127	DispatchNorth	Bethpage Di...	Bethpage Di...	Dist Ops	Local Lineman	No	Chris B Hallett
☐	1107	DispatchNorth	Local Coho ...	Local Coho ...	Dist Ops	Local Lineman	No	Scott A Cooper
☐	1105	DispatchNorth	Local Coho ...	Local Coho ...	Dist Ops	Local Lineman	No	Darrin G Bowl
☐	1106	DispatchNorth	Local Coho ...	Local Coho ...	Dist Ops	Local Lineman	No	Russell Ben Brobak
☐	1108	DispatchNorth	Local Coho ...	Local Coho ...	Dist Ops	Local Lineman	No	William X Larive
☐	1131	DispatchNorth	Muirfield Dis...	Muirfield Dis...	Dist Ops	Local Lineman	No	Mike Jeff Hackett
☐	1129	DispatchNorth	Muirfield Dis...	Muirfield Dis...	Dist Ops	Local Lineman	No	Adam D Johnson

Figure 6. Crews Tab

The Crews tab shows the following information:

- **Select:** Select this check box to select the crew record.
- **Radio Number:** Crew's radio number.
- **Crew Division:** Name of the division the crew belongs to.
- **Crew Area:** The current crew area name.
- **Default Crew Area:** The default crew area name.
- **Department:** Name of the department the crew belongs to.
- **Crew Type:** Crew type description.
- **Is Evaluator:** If yes, the crew primary employee is evaluator.
- **Primary Emp:** Name of the crew's primary employee.
- **#Emp:** Employee count for this crew.

- **#Vehicle:** Vehicle count for this crew.
- **MDT Flag:** If True, the crew is equipped with a mobile data terminal.
- **Phone Num:** The crew phone number.
- **Temp Area:** Area the crew is temporarily assigned to.
- **Auto Dispatch:** Indicates if the crew is enabled for the auto dispatch process.

2.4.1. Update Crew Pane

Click on a crew record in the Crew Status grid to open the Update Crew pane populated with the selected crew record data.

Crew Modeling

File Help

Update Crew

Properties

Name	Value
Auto Dispatch	True
Automatic Dispatch Id	91101
Company Name	
Contractor	False
Crew Area*	3Lions District Crews
Crew Division*	DispatchNorth
Crew Type	Local Lineman
Default Crew Area*	3Lions District Crews
Department	Dist Ops
Evaluator	False
Foreign	False
Home Area	3Lions District Crews

Primary ... Tony Duchesneau Make This Emplov...

Current crew employees

ID	NAME	TEMPCREWNAME	RADIONUMBER

Filter:
↑ ↓

Available employees

ID	NAME	CREWNAME	TEM
FI2685	Bonnie L Persons		
FI1502	Daniel Jeffrey Dodge	Daniel Jeffrey Dodge	
FI1571	Ethan G Bertilson	Ethan G Bertilson	

Employees Vehicles

Cancel OK

Figure 7. Update Crew Panel of the Crew Management Display

The Update Crew pane contains the following buttons:

- **Cancel:** Click to discard the changes and close the Update Crew pane.
- **OK:** Click to save the changes.

2.4.1.1. Update Crew Pane - Properties Pane

The Properties pane contains the following fields:

- **Auto Dispatch:** Controls the crews for auto dispatch process. This is used only for the Process Automation application. The default value is set to True.
- **Automatic Dispatch ID:** The unique identification number from the Work Force Management system to track the crew's coordinates and communicate the crew location to the OMS system.
- **Company Name:** Name of the company the foreign crew belongs to (this field is enabled only when the Foreign flag is True).
- **Contractor:** Select this check box if the crew is a contractor.
- **Crew Area:** The current crew area name.
- **Crew Division:** The division the crew operates within.
- **Crew Type:** The crew type (Foreman, Lineman etc.).
- **Default Crew Area:** The default crew area name.
- **Department:** Name of the department the crew belongs to.
- **Evaluator:** Flag to indicate if the crew is the evaluator.
- **Foreign:** If set to True, the crew reports to another company.
- **ID:** Crew ID.
- **MDT:** If True, the crew is equipped with a mobile data terminal.

Important

To enable integration with the Crew Server web interface and Workforce Management System, the MDT field must be set to True and the Automatic Dispatch ID field must be populated for primary employees of crews that are equipped with a mobile data terminal (MDT).

- **Name:** Name of the crew.
- **Phone Number:** The crew's phone number.
- **Primary Employee ID:** Identification number of the crew's primary employee.
- **Primary Employee Name:** Name of the crew's primary employee.
- **Radio Number:** Crew's radio number.
- **Temporary:** If True, the crew is temporary (call out).

Note

In case of a storm situation, foreign crews may be required to restore the power. A non-company crew must have a regular company employee attached to it as a main contact person.

2.4.1.2. Crew Pane - Employees Tab

Primary Employee: Craig D Hutchinson Make This Employee Primary

Current crew employees

ID	NAME	TEMPCREWNAME	RADIONUMBER	CRE
FI2081	Robert Steve Tyson		4191	Cent
FI2079	Robert T. Crowley		4190	Cent

Filter: ↑ ↓

Available employees

ID	NAME	CREWNAME	TEMPCREWNAME
FI1826	Bill Z Stelzner	Brent N Phillips	
FI1849	Bradley T Resh	N Alvin Lewis	
FI1881	Bradley T Thomas		
FI2007	Brent Darrin Shearer	Brent Darrin Shearer	

Employees Vehicles

Figure 8. Employees Tab of the Update Crew Panel

The Employees tab contains the following information:

- **Primary Employee:** Name of the crew's primary employee.
- **Make This Employee Primary:** This button is enabled only for temporary crews (refer to section [Creating a Temporary Crew](#)).
- **Current Crew Employees:** Shows employees associated with the selected crew.
- **Filter:** Allows you to filter the list of employees by name.
- **Up Arrow Button:** Click to add the employee selected in the Available Employees box to the Current Crew Employees.
- **Down Arrow Button:** Click to remove the selected employee from the Current Crew Employees.
- **Available Employees Box:** The Available Employees box lists all employees associated with the selected crew area that can be added to the crew.

2.4.1.3. Crew Pane - Vehicles Tab

Crew employees

ID	NAME	VEHICLEIDS
FI1857	Aaron Tad Murphy	
FI2075	Ben O Myers	
FI1862	James Jay Druhan	

Crew vehicles

Id	Name	VehicleTypeId	EmployeeId	RadioNumber
VH20124	Bucket truck 20124	Bucket truck	FI1943	

Filter:

 Select the crew member first then the vehicle.

Vehicles

Id	Name	VehicleTypeId	EmployeeId	RadioNumber
VH100...	Pickup 10000	Pickup		
VH100...	Pickup 10001	Pickup		
VH100...	Pickup 10002	Pickup		
VH100...	Pickup 10003	Pickup		

Figure 9. Vehicles Tab of the Update Crew Panel

The Vehicles tab contains the following information:

- **Crew Employees Box:** Shows a list of the crew's employees and vehicles assigned to each employee.
- **Crew Vehicles Box:** Shows a list of vehicles assigned to the selected crew.
- **Filter:** Allows you to filter the list of available vehicles by name.
- **Up Arrow Button:** Click to assign the selected vehicle from the available vehicles list to the selected employee.
- **Down Arrow Button:** Click to de-assign the selected vehicle from the crew.
- **Vehicles Box:** Shows the list of all available vehicles that can be assigned to the crew.

2.5. Employee Types Tab

Use the Employee Types tab to view, add, and update employee type records.

Employees	Crews	Employee Types	Crew Areas	Vehicle Types	Vehicles	Messages
Employee Types: 						
Employee Types   						
☐	Id	Name	AutoGenerateCrew	DoNotImplement	Is Evaluator	
☐	1100	Administrator	No	No	No	
☐	1105	Administrator, Sr.	No	No	No	
☐	1110	Business Office Assistant	No	No	No	
☐	1300	Business Office Specialist	No	No	No	
☐	1501	Business Office Manager	No	No	No	
☐	2001	Utility Assistant	No	No	No	
☐	2005	Appr Cable Splicer	No	No	No	
☐	2010	Cable Splicer	No	No	No	
☐	2015	Lead Cable Splicer	Yes	No	No	
☐	2055	Apprentice Lineman	No	No	No	
☐	2060	Lineman	No	No	No	
☐	2065	Lead Lineman	Yes	No	No	
☐	3100	Local Lineman	Yes	No	No	
☐	4005	Operations Assistant I	No	No	No	
☐	4010	Operations Assistant II	No	No	No	
☐	4012	Operations Assistant III	No	No	No	
☐	4015	Operations Assistant, Sr.	No	No	No	
☐	5000	Distribution Specialist	Yes	No	Yes	
☐	5005	Distribution Specialist, Sr.	Yes	No	Yes	

Figure 10. Employee Types Tab

The Employee Types tab shows the following information:

- **Select:** Select this check box to select the record.
- **ID:** Employee type ID.
- **Name:** Name of the employee type.
- **AutoGenerateCrew:** Flag to determine if the employees with the given employee type should have their own crew. If the flag is set to True, ADMS automatically creates a crew with the given employee as its primary employee. The default value is False.
- **DoNotImplement:** If set to Yes, prevents creating a crew for this employee type. This could be used to temporarily not create a crew for this employee type.
- **Is Evaluator:** If set to Yes, this employee type is an evaluator.

2.5.1. Employee Type Tab Context Menu

You can access this context menu by right-clicking on any employee record.

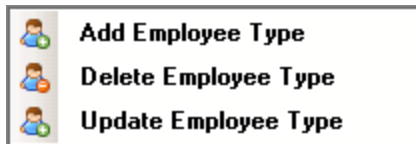


Figure 11. Employee Type Tab Context Menu

The following options are available:

- **Add Employee Type:** Activates the Add Employee Type pane, where you can create a record for a new employee type.
- **Delete Employee Type:** Deletes the selected employee type record.
- **Update Employee Type:** Activates the Update Employee Type pane, where you can update the selected employee type record.

When several records are selected, the context menu options include:

- **Delete Employee Types:** Deletes all selected employee types.
- **Update Employee Types:** Activates the Update Employee Type pane, where you can update the selected employee type records.

2.5.2. Add/Update Employee Type Pane

Name	Value
Auto Generate Crew	False
Do Not Implement	False
Id	1110
Is Evaluator	False
Name	Business Office Assistant

Figure 12. Update Employee Type Pane

Use the **Add Employee Type** pane to create a new employee type record. To display this pane, select the Add Employee option on the Employee Type tab's context menu.

Use the **Update Employee Type** pane to change the selected employee type information. To display this pane, select the Update Employee Type option on the context menu or click on the desired employee record.

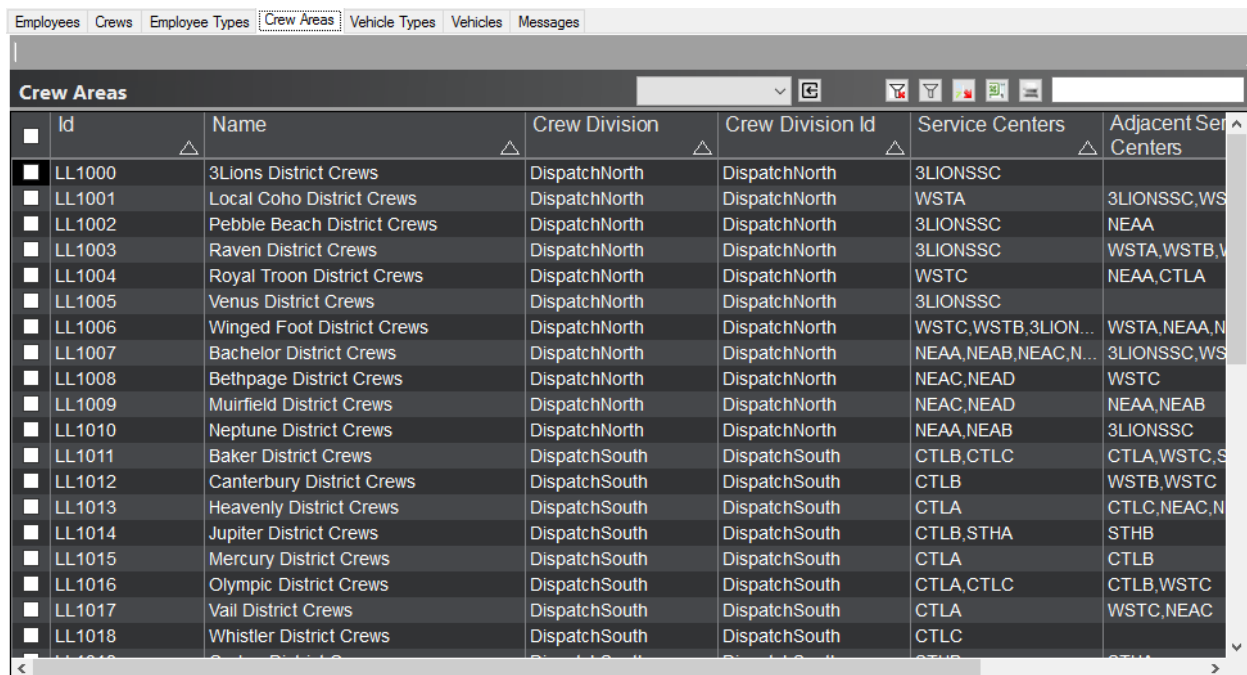
The Add/Update Employee Type pane includes the following fields:

- **Auto Generate Crew:** Determines if the employee with the given employee type should have a crew. If the flag is set to True, ADMS automatically creates a crew with the given employee as its primary employee. The default value is False.
- **Do Not Implement:** If False, prevents creating a crew for this employee type. This could be used to temporarily not create a crew for this employee type.

- **ID:** The employee type identification number.
- **Is Evaluator:** If set to True, this employee type is the evaluator.
- **Name:** Name of the employee type.
- **Cancel:** Click to discard the changes and close the Add/Update pane.
- **OK:** Click to save the changes.

2.6. Crew Areas Tab

Use the Crew Areas tab to view, add, and update crew area records.



	Id	Name	Crew Division	Crew Division Id	Service Centers	Adjacent Ser Centers
☐	LL1000	3Lions District Crews	DispatchNorth	DispatchNorth	3LIONSSC	
☐	LL1001	Local Coho District Crews	DispatchNorth	DispatchNorth	WSTA	3LIONSSC, WS
☐	LL1002	Pebble Beach District Crews	DispatchNorth	DispatchNorth	3LIONSSC	NEAA
☐	LL1003	Raven District Crews	DispatchNorth	DispatchNorth	3LIONSSC	WSTA, WSTB, V
☐	LL1004	Royal Troon District Crews	DispatchNorth	DispatchNorth	WSTC	NEAA, CTLA
☐	LL1005	Venus District Crews	DispatchNorth	DispatchNorth	3LIONSSC	
☐	LL1006	Winged Foot District Crews	DispatchNorth	DispatchNorth	WSTC, WSTB, 3LION...	WSTA, NEAA, N
☐	LL1007	Bachelor District Crews	DispatchNorth	DispatchNorth	NEAA, NEAB, NEAC, N...	3LIONSSC, WS
☐	LL1008	Bethpage District Crews	DispatchNorth	DispatchNorth	NEAC, NEAD	WSTC
☐	LL1009	Muirfield District Crews	DispatchNorth	DispatchNorth	NEAC, NEAD	NEAA, NEAB
☐	LL1010	Neptune District Crews	DispatchNorth	DispatchNorth	NEAA, NEAB	3LIONSSC
☐	LL1011	Baker District Crews	DispatchSouth	DispatchSouth	CTLB, CTLC	CTLA, WSTC, S
☐	LL1012	Canterbury District Crews	DispatchSouth	DispatchSouth	CTLB	WSTB, WSTC
☐	LL1013	Heavenly District Crews	DispatchSouth	DispatchSouth	CTLA	CTLC, NEAC, N
☐	LL1014	Jupiter District Crews	DispatchSouth	DispatchSouth	CTLB, STHA	STHB
☐	LL1015	Mercury District Crews	DispatchSouth	DispatchSouth	CTLA	CTLB
☐	LL1016	Olympic District Crews	DispatchSouth	DispatchSouth	CTLA, CTLC	CTLB, WSTC
☐	LL1017	Vail District Crews	DispatchSouth	DispatchSouth	CTLA	WSTC, NEAC
☐	LL1018	Whistler District Crews	DispatchSouth	DispatchSouth	CTLC	

Figure 13. Crew Areas Tab

The Crew Areas tab shows the following information:

- **Select:** Select this check box to select the record.
- **ID:** Crew area ID.
- **Name:** Name of the crew area.
- **Crew Division:** Name of the crew division the crew area belongs to.
- **Crew Division ID:** Identification number of the crew division the crew area belongs to.
- **Service Centers:** ID of the service center of the crew area.

Service centers are logical and/or geographic groupings of customers. They are modelled in the customer model (refer to the *Customer List Compiler User Guide*).

Crew areas may be associated with one or more service centers. To associate several service

centers with a crew area, enter the IDs separated by a comma.

- **Adjacent Service Centers:** Neighboring service centers. This field is also used for filtering crew areas. When the Adjacent Service Centers field is specified, the neighboring crew areas are included in the filtering result by default.
- **Truck Zones:** ID of the truck zone of the crew area.

Crew areas may be associated with one or more truck zones. To associate several truck zones with a crew area, enter the IDs separated by a comma.

- **X/Y:** X and Y coordinates of the crew area on the geographic network view.

2.6.1. Crew Areas Tab Context Menu

You can access this context menu by right-clicking on any record.

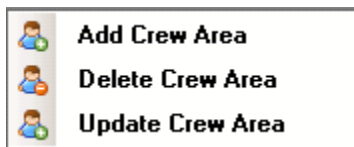


Figure 14. Crew Areas Tab Context Menu

The following options are available:

- **Add Crew Area:** Activates the Add Crew Area pane, where you can create a record for a new crew area.
- **Delete Crew Area:** Deletes the selected crew area record.
- **Update Crew Area:** Activates the Update Crew Area pane, where you can update the selected crew area record.

When several records are selected, the context menu options include:

- **Delete Crew Areas:** Deletes all selected crew areas.
- **Update Crew Areas:** Activates the Update Crew Area pane, where you can update the selected crew areas.

2.6.2. Add/Update Crew Area Pane

Use the **Add Crew Area** pane to create a new crew area record. To display this pane, select the Add Crew Area option on the Crew Areas tab's context menu.

Use the **Update Crew Area** pane to change the selected crew area information. To display this pane, select the Update Crew Area option on the context menu or click on the desired crew area.

Update Crew Area

Properties

Name	Value
Adjacent Service Centers	3LIONSSC
Crew Division	DispatchNorth
Crew Division Id	DispatchNorth
Id	LL1010
Location X	959998
Location Y	818331
Name	Neptune District Crews
Service Centers	NEAA,NEAB

Cancel
OK

Figure 15. Update Crew Area Pane

The Add/Update Crew Area pane shows the following information:

- **Adjacent Service Centers:** Name of the adjacent service centers to the crew area service centers.
- **Crew Division:** Name of the crew division that the crew area belongs to.
- **Crew Division ID:** Identification number of the crew division that the crew area belongs to.
- **ID:** Identification number of the crew area.
- **Location X/Y:** X and Y coordinates of the crew area on the geographic network view.
- **Name:** Name of the crew area.
- **Service Centers:** Name of the service centers the crew area is associated with.

Service centers are logical and geographic groupings of customers. They are modelled in the

customer model (refer to the *Customer List Compiler User Guide*).

- **Cancel:** Click to discard the changes and close the Add/Update pane.
- **OK:** Click to save the changes.

2.7. Vehicle Types Tab

Use the Vehicle Types tab to view, add, and update vehicle type records.

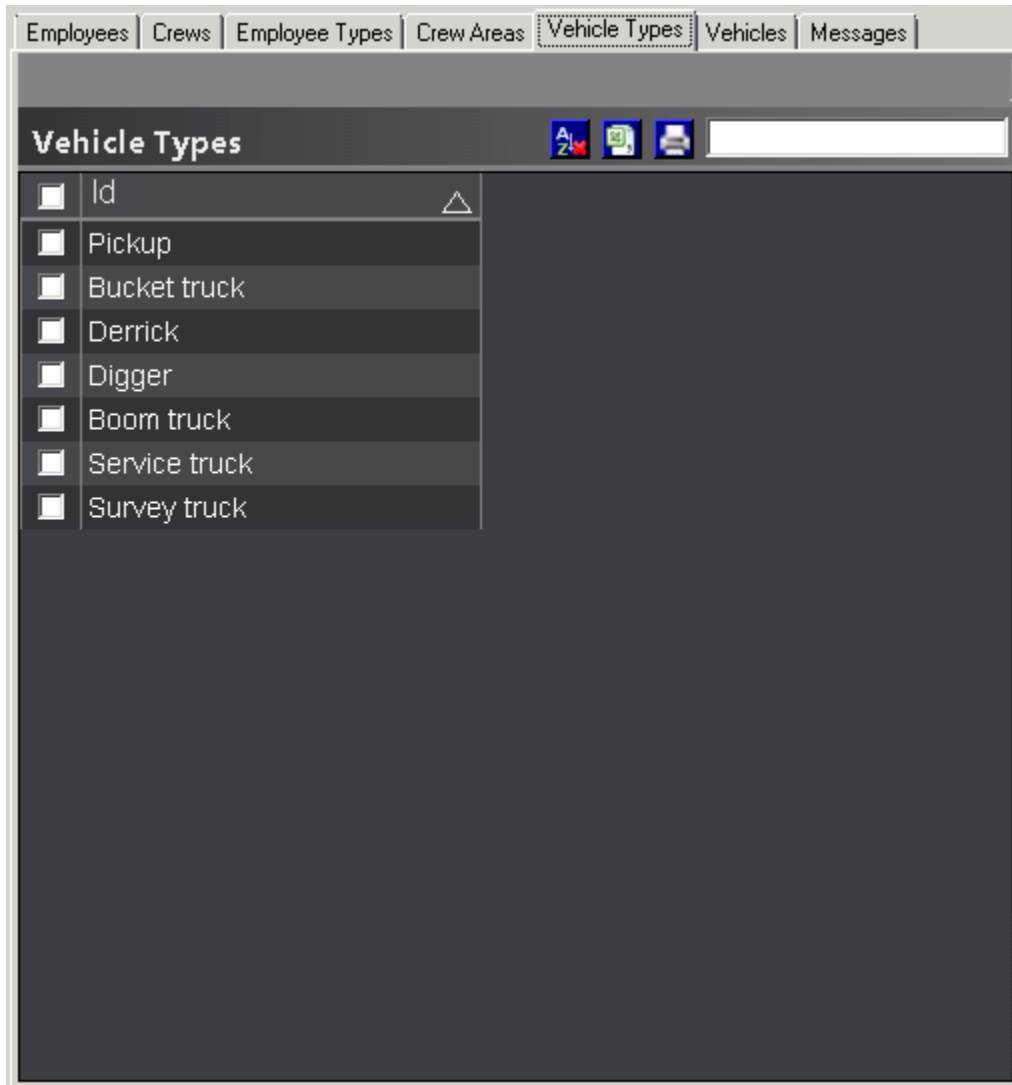


Figure 16. Vehicle Types Tab

The Vehicle Types tab shows the following information:

- **Select:** Select this check box to select the record.
- **ID:** Identification name of the vehicle type.

2.7.1. Vehicle Types Tab Context Menu

You can access this context menu by right-clicking on any record.

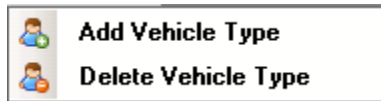


Figure 17. Vehicle Types Tab Context Menu

The following options are available:

- **Add Vehicle Type:** Activates the Add Vehicle Types pane, where you can create a record for a new vehicle type.
- **Delete Vehicle Type:** Deletes the selected record.

When several records are selected, the context menu options include:

- **Delete Vehicle Types:** Deletes all selected vehicle types.

2.7.2. Add/Update Vehicle Type Pane

Use the **Add Vehicle Type** pane to create a new vehicle type. To display this pane, select the Add Vehicle Type option on the Vehicle Types tab's context menu.

Use the **Update Vehicle Type** pane to change the selected record information. To display this pane, click on the desired record on the Vehicle Types tab.

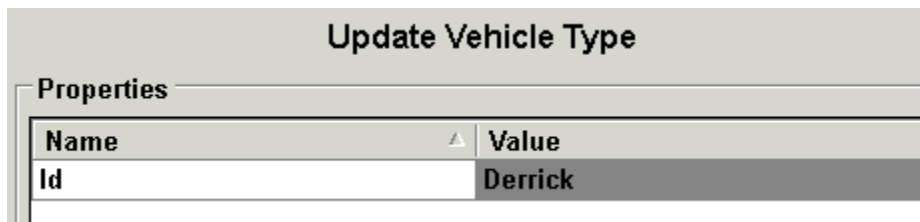


Figure 18. Update Vehicle Type Pane

The Add/Update Vehicle Type pane shows the following information:

- **ID:** Identification name of the vehicle type.
- **Cancel:** Click to discard the changes and close the Add/Update pane.
- **OK:** Click to save the changes.

2.8. Vehicles Tab

Use the Vehicles tab to view, add, and update vehicle records.

Employees Crews Employee Types Crew Areas Vehicle Types Vehicles Messages					
Vehicles					
☐	Id	Name	Vehicle Type Id	Employee Id	Rad
☐	VH20125	Bucket truck 20125	Bucket truck	FI1944	
☐	VH20126	Bucket truck 20126	Bucket truck	FI1945	
☐	VH20127	Bucket truck 20127	Bucket truck	FI1946	
☐	VH20128	Bucket truck 20128	Bucket truck	FI1947	
☐	VH20129	Bucket truck 20129	Bucket truck	FI1948	
☐	VH20130	Bucket truck 20130	Bucket truck	FI1949	
☐	VH20131	Bucket truck 20131	Bucket truck	FI1950	
☐	VH20132	Bucket truck 20132	Bucket truck	FI1951	
☐	VH20133	Bucket truck 20133	Bucket truck	FI1952	
☐	VH20134	Bucket truck 20134	Bucket truck	FI1953	
☐	VH20135	Bucket truck 20135	Bucket truck	FI1954	
☐	VH20136	Bucket truck 20136	Bucket truck	FI1955	
☐	VH20137	Bucket truck 20137	Bucket truck	FI1956	
☐	VH20138	Bucket truck 20138	Bucket truck	FI1957	

Figure 19. Vehicles Tab

The Vehicles tab shows the following information:

- **Select:** Select this check box to select the record.
- **ID:** Identification name of the vehicle.
- **Name:** Descriptive name of the vehicle.
- **Vehicle Type ID:** Identification name of the vehicle type.
- **Employee ID:** Identification number of the employee the vehicle is assigned to. By specifying the employee ID, this vehicle is associated with the crew that the employee belongs to. This can be used as an indication of the crew's capabilities.
- **Radio Number:** Radio number associated with the vehicle. Use this field if the vehicle's radio number is different than the employee's radio number (for example, when a radio is installed in the vehicle).
- **Phone Number:** Phone number associated with the vehicle.
- **Crew ID:** Crew ID associated with the vehicle.
- **Reported Time:** The time when the most recent location update was received from the vehicle.

2.8.1. Vehicle Tab Context Menu

You can access this context menu by right-clicking on any record.

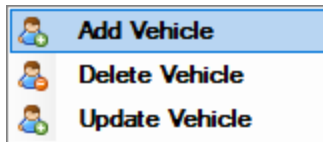


Figure 20. Vehicle Tab Context Menu

The following options are available:

- **Add Vehicle:** Activates the Add Vehicle pane, where you can create a record for a new vehicle.
- **Update Vehicle:** Activates the Update Vehicle pane, where you can update a vehicle record.
- **Delete Vehicle:** Deletes the selected record.

When several records are selected, the context menu options include:

- **Delete Vehicles:** Deletes all selected vehicles.
- **Update Vehicle:** Activates the Update Vehicle pane, where you can update a vehicle record.

2.8.2. Add/Update Vehicle Pane

Use the **Add Vehicle** pane to create a new vehicle type. To display this pane, select the Add Vehicle option on the Vehicle tab's context menu.

Use the **Update Vehicle** pane to change the selected vehicle record. To display this pane, select the Update Vehicle option on the Vehicle tab's context menu or click on the desired record.

Update Vehicle

Properties

Name	Value
Crew Id	<None> ▼
Employee Id	FI1500 ▼
Id	VH20000
Name	Bucket truck 20000
Phone Number	
Radio Number	
Vehicle Type Id	Bucket truck ▼

Cancel

OK

Figure 21. Update Vehicle Pane

The Add/Update Vehicle pane shows the following information:

- **Crew ID:** Crew ID associated with the vehicle.
- **Employee ID:** Identification number of the employee that the vehicle is assigned to. By specifying the employee ID, this vehicle is associated with the crew that the employee belongs to. This can be used as an indication of the crew's capabilities.
- **ID:** Identification number of the vehicle.
- **Name:** Vehicle name.
- **Phone Number:** Phone number associated with the vehicle.
- **Radio Number:** Radio number associated with the vehicle. Use this field if the vehicle's radio number is different than the employee's radio number (for example, when a radio is installed in the vehicle).
- **Vehicle Type ID:** Identification name of the vehicle type.
- **Cancel:** Click to discard the changes and close the Add/Update pane.

- **OK:** Click to save the changes.

2.9. Messages Tab

The Messages tab shows success and error messages for all crew actions.

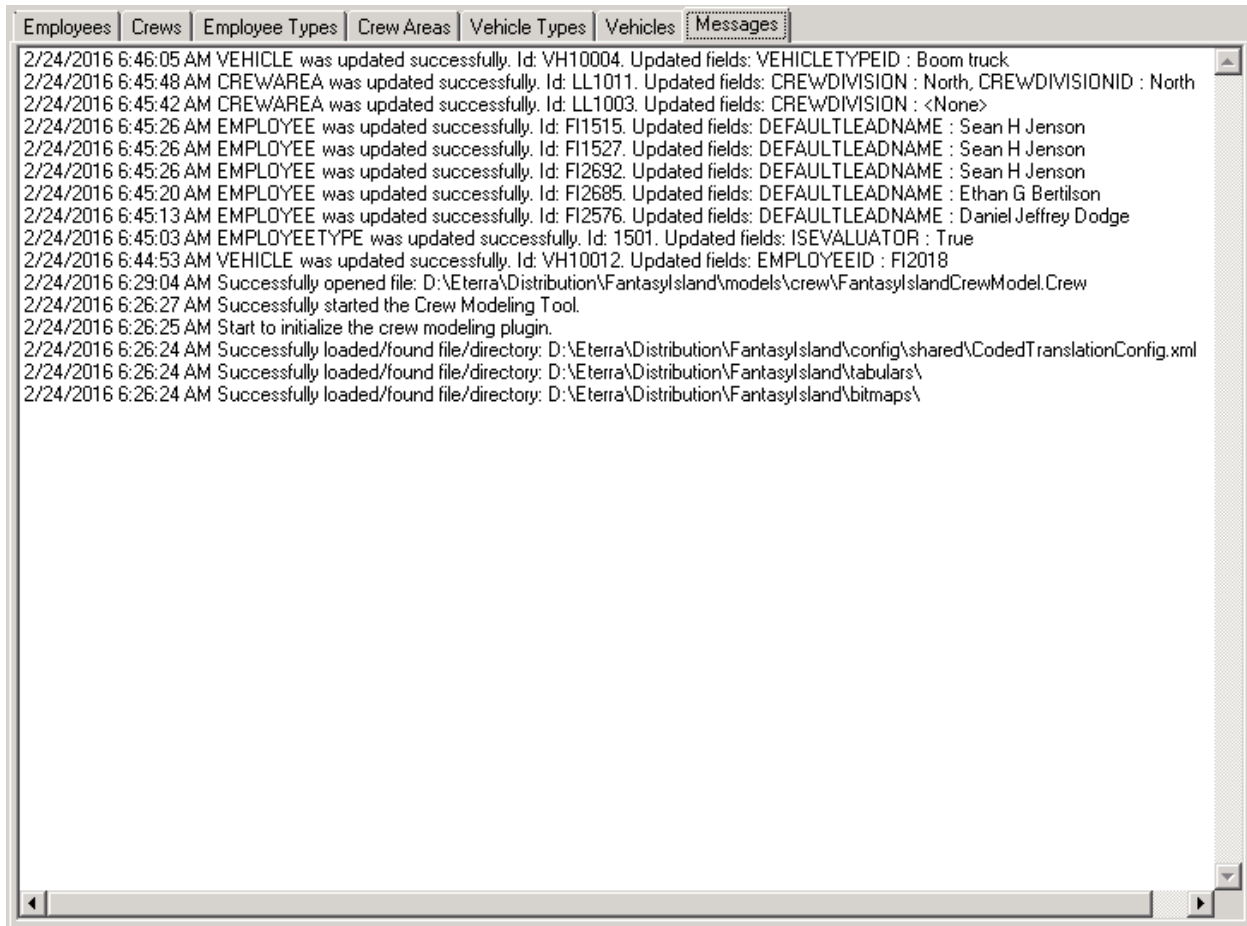


Figure 22. Messages Tab

3. Using the Crew Modeling Application

This chapter describes the crew modeling displays and menu options, and provides instructions about how to access and operate the Crew Modeling application.

3.1. Starting the Crew Modeling Application

To start the Crew Modeling application:

- From the Microsoft Windows Start menu, navigate to All Programs > Eterra > Distribution > Tools > Crew Modeling Tool.

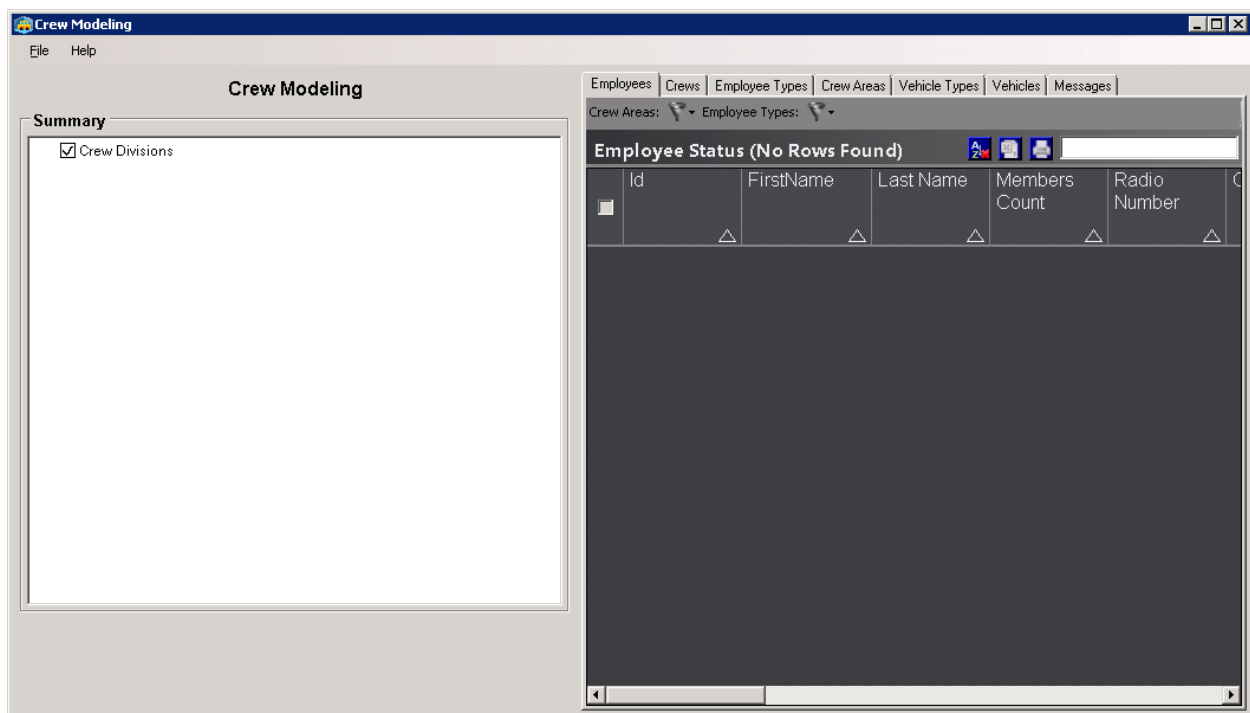


Figure 23. Crew Modeling Application

The Crew Modeling tool is a standalone Windows application that allows manual entry of static crew data and importing and exporting crew data.

3.2. Managing Crews

You can add, delete, and maintain the crew and employee data using the Crew Modeling application.

3.2.1. Updating a Crew

To update a selected crew record:

1. Select a crew record in the Crew Status grid.

The Update Crew pane appears, populated with the selected crew record data.

2. Change the crew information.

The screenshot shows the 'Update Crew' pane on the left and the 'Crew Status' grid on the right. The 'Update Crew' pane has a 'Properties' section with fields for Name, Automatic Dispatch Id (11102), Company Name, Contractor (False), Crew Area (3Lions District Crews), Crew Division (DispatchNorth), Crew Type (Local Lineman), and Default Crew Area (3Lions District Crews). Below this is the 'Primary Employee' section with 'Sam A Sheridan' and a 'Make This Employee Primary' button. The 'Current crew employees' section has a table with columns ID, NAME, TEMPCREWNAME, RADIONUMBER, and CREWAREA. The 'Available employees' section has a table with columns ID, NAME, CREWNAME, and TEMPCREWNAME. At the bottom are 'Cancel' and 'OK' buttons. The 'Crew Status' grid on the right has columns for Radio Num, Crew Divisor, Crew Area, Default Crew Area, Department, and Crew. It contains a list of crew records with checkboxes in the first column.

3. Click OK to save the changes.

The crew record is now updated.

3.2.1.1. Adding and Removing Crew Members

To add a member to a crew:

1. On the Update Crew pane, select the employee in the box containing the list of available employees.
2. Click the Up arrow button. The employee is moved from the Available Employees box to the Current Crew Employees box (under the primary employee name).

Primary Employee: Daniel Jeffrey Dodge Make This Employee Primary

Current crew employees

ID	NAME	TEMPCREWNAME	RADIONUMBER	CREWARE
FI2685	Bonnie L Persons			3Lions Dist

Filter: ↑ ↓

Available employees

ID	NAME	CREWNAME	TEMPCREWNAME	RADIC
FI1571	Ethan G Bertilson	Ethan G Bertilson		1100
FI2576	Mary I Crosby			
FI1564	Ron X Stanton	Ron X Stanton		1103
FI2625	Ryan A Gorman			

Employees Vehicles

To remove a member from a crew:

1. Select the employee in the Current Crew Employees box.
2. Click the Down arrow button. The employee is moved to the Available Employees box.

3.2.1.2. Assigning and De-assigning Vehicles

To assign a vehicle to a crew:

1. Click the Vehicle tab on the Update Crew pane.

The Current Crew Employees (top box) is populated with the list of the crew's employees and vehicles assigned to each employee.

The Vehicle per Crew box (middle box) is populated with the list of vehicles currently assigned to the selected crew.

2. Select the employee in the Current Crew Employees (top box).
3. Select a vehicle from the Available Vehicles box (bottom box) that you need to assign to the selected crew employee. The Up arrow button is enabled.

Crew employees

ID	NAME	VEHICLEIDS
FI2685	Bonnie L Persons	
FI1502	Daniel Jeffrey Dodge	VH20002

Crew vehicles

Id	Name	VehicleTypeId	EmployeeId
VH20002	Bucket truck 20002	Bucket truck	FI1502

Filter:   Select the crew member first then the vehicle.

Vehicles

Id	Name	VehicleTypeId	EmployeeId
VH10000	Pickup 10000	Pickup	
VH10001	Pickup 10001	Pickup	

Employees Vehicles

- Click the Up arrow button to assign the selected vehicle to the employee. The vehicle is moved from the Available Vehicles box to the Current Crew Employees and the Vehicle per Crew box.

Crew employees

ID	NAME	VEHICLEIDS
FI2685	Bonnie L Persons	VH10000
FI1502	Daniel Jeffrey Dodge	VH20002

Crew vehicles

Id	Name	VehicleTypeId	EmployeeId
VH20002	Bucket truck 20002	Bucket truck	FI1502
VH10000	Pickup 10000	Pickup	FI2685

Filter:   Select the crew member first then the vehicle.

Vehicles

Id	Name	VehicleTypeId	EmployeeId
VH10001	Pickup 10001	Pickup	
VH10002	Pickup 10002	Pickup	

Employees Vehicles

- Click OK.

To de-assign a vehicle from a crew:

1. Click the Vehicle tab on the Update Crew pane.

The Current Crew Employees (top box) is populated with the list of the crew's employees and vehicles assigned to each employee.

The Vehicle per Crew box (middle box) is populated with the list of vehicles currently assigned to the selected crew.

2. Select a vehicle from the Vehicle per Crew box (middle box) that you need to de-assign from the crew. The Down arrow button is enabled.

Crew employees

ID	NAME	VEHICLEIDS
FI2685	Bonnie L Persons	VH10000
FI1502	Daniel Jeffrey Dodge	VH20002

Crew vehicles

Id	Name	VehicleTypeId	EmployeeId
VH20002	Bucket truck 20002	Bucket truck	FI1502
VH10000	Pickup 10000	Pickup	FI2685

Filter: Select the crew member first then the vehicle.

Vehicles

Id	Name	VehicleTypeId	EmployeeId
VH10001	Pickup 10001	Pickup	
VH10002	Pickup 10002	Pickup	

Employees Vehicles

3. Click the Down arrow button. The vehicle is now de-assigned from the crew and added to the Available Vehicles box (bottom box).
4. Click OK.

3.2.2. Splitting a Crew

The Split Crew option is available only from the OMS Crews tab of the OMS Display Plug-in. This option allows you to create a new temporary crew from one or more of the employees of an existing crew. This option is only available if the crew consists of more than one member.

To split a crew:

1. In the OMS Display Plug-In, on the Crews tab, right-click the desired crew record and select the Update/Split/Create Temp Crew option.

The Update Crew dialog box appears, populated with the selected crew record data.

- On the Employees tab, select the employee in the Current Crew Employees box under the primary employee name. The Split Crew button is now enabled.

Crew Update Dialog - Crew Name: Carino (1207)

Properties

Name	Value
Active	True
Automatic Dispatch Id	11207
Company Name	
Contractor	False
Crew Area	N Jupiter District Crews
Crew Division	South
Crew Status	SignOn
Crew Type	Local Lineman
Default Location X	916556
Default Location Y	751871
Department	Dist Ops
Evaluator	False
Foreign	False
Id	FI1500
MDT	False
Name	Carino (1207)
Phone Number	555-129-5174
Primary Employee Id	FI1500
Primary Employee Name	Q Jeff Carino
Radio Number	1207
Temporary	False

Primary Employee: Q Jeff Carino [Make This Employee Primary](#) [Create Temp Crew](#)

Current crew employees

ID	NAME	TEMPCREWNAME	RADIONUMBER	CREWAREA	DEFAULTCREWAREA	EMPLOYEEYPENAME
FI1587	Ethan C Barber		1206	N Jupiter District Crews	N Jupiter District Crews	Local Lineman

Filter: [Split Crew](#)

Available employees

ID	NAME	CREWNAME	TEMPCREWNAME	RADIONUMBER	CREWAREA	DEFAULTCREWAREA	EMPLC
FI1555	Jim Chris Carino	Carino (1208)		1208	N Jupiter District Crews	N Jupiter District Crews	Local L
FI1601	Jon C Joiner	Joiner (1210)		1210	N Jupiter District Crews	N Jupiter District Crews	Local L
FI1560	K Geoff Gowdy	Gowdy (1209)		1209	N Jupiter District Crews	N Jupiter District Crews	Local L
FI2720	Kathy D Brobak				N Jupiter District Crews	N Jupiter District Crews	Manag

Employees Vehicles

OK Cancel Apply

- Click Split Crew. A temporary crew record appears on the Properties pane and the selected crew member from the previous step is the primary employee.

Crew Update Dialog - Crew Name: Barber [TEMPFI1587]

Properties

Name	Value
Active	True
Automatic Dispatch Id	
Company Name	
Contractor	False
Crew Area	N Jupiter District Crews
Crew Division	South
Crew Status	SignOn
Crew Type	Local Lineman
Default Location X	0
Default Location Y	0
Department	
Evaluator	False
Foreign	False
Id	TEMPFI1587
MDT	False
Name	Barber [TEMPFI1587]
Phone Number	
Primary Employee Id	FI1587
Primary Employee Name	Ethan C Barber
Radio Number	1206
Temporary	True

Primary Employee: Ethan C Barber [Make This Employee Primary](#) [Create Temp Crew](#)

Current crew employees

ID	NAME	TEMPCREWNAME	RADIONUMBER	CREWAREA	DEFAULTCREWAREA	EMPLOYEEYPENAME	MEMBERSC

Filter: [Split Crew](#)

Available employees

ID	NAME	CREWNAME	TEMPCREWNAME	RADIONUMBER	CREWAREA	DEFAULTCREWAREA	EMPLC
FI1555	Jim Chris Carino	Carino (1208)		1208	N Jupiter District Crews	N Jupiter District Crews	Local L
FI1601	Jon C Joiner	Joiner (1210)		1210	N Jupiter District Crews	N Jupiter District Crews	Local L
FI1560	K Geoff Gowdy	Gowdy (1209)		1209	N Jupiter District Crews	N Jupiter District Crews	Local L
FI2720	Kathy D Brobak				N Jupiter District Crews	N Jupiter District Crews	Manag
FI1500	Q Jeff Carino	Carino (1207)		1207	N Jupiter District Crews	N Jupiter District Crews	Local L

Employees Vehicles

OK Cancel Apply

If a primary employee has team members, the number of members is shown near the primary employee name. This is useful for foreign (mutual aid) employees.

Crew Update Dialog - Crew Name: McKeag [TEMPFI2797]

Properties

Name	Value
Active	True
Automatic Dispatch Id	
Company Name	
Contractor	False
Crew Area	North Division Office
Crew Division	North
Crew Status	SignOn
Crew Type	Business Office Supervisor
Default Location X	0
Default Location Y	0
Department	
Evaluator	False
Foreign	False
Id	TEMPFI2797
MDT	False
Name	McKeag [TEMPFI2797]
Phone Number	
Primary Employee Id	FI2797
Primary Employee Name	Lillian M McKeag
Radio Number	

Primary Employee: Lillian M McKeag - with 10... [Make This Employee Primary](#) [Create Temp Crew](#)

Current crew employees

ID	NAME	TEMPCREWNAME	RADIONUMBER	CREWAREA	DEFAULTCREWAREA	EMPLOYEEYTYPE
----	------	--------------	-------------	----------	-----------------	---------------

Filter: [Split Crew](#)

Available employees

ID	NAME	CREWNAME	TEMPCREWNAME	RADIONUMBER	CREWAREA
FI2018	A Ben Druhan	Jackson [TEMPFI1808]	Jackson [TEMPFI1808]	1407	West Crews
FI1950	A Paul Morgan	Cannon [TEMPFI1800]	Cannon [TEMPFI1800]	4264	Central Crews
FI1918	A Rusty Connell	Jackson [TEMPFI1808]	Jackson [TEMPFI1808]	4048	South Crews
FI2718	Aaron Craig Cowley	Jackson [TEMPFI1808]	Jackson [TEMPFI1808]		Coast Pinehur
FI2141	Aaron P Dorrell	Cannon [TEMPFI1800]	Cannon [TEMPFI1800]	4156	West Crews
FI1607	Aaron P Drost	Drost (1135)		1135	Neptune Distr
FI1857	Aaron Tad Murphy	Crawford [5251] - AUTO		4085	West Crews
FI1841	Aaron X Cappucci	Thompson [5274] - AUTO		4104	Northeast Cre

[Employees](#) [Vehicles](#) [OK](#) [Cancel](#) [Apply](#)

4. Add more employees to the crew if needed.

Crew Update Dialog - Crew Name: Barber [TEMPFI1587]

Properties

Name	Value
Active	True
Automatic Dispatch Id	
Company Name	
Contractor	False
Crew Area	N Jupiter District Crews
Crew Division	South
Crew Status	SignOn
Crew Type	Local Lineman
Default Location X	0
Default Location Y	0
Department	
Evaluator	False
Foreign	False
Id	TEMPFI1587
MDT	False
Name	Barber [TEMPFI1587]
Phone Number	
Primary Employee Id	FI1587
Primary Employee Name	Ethan C Barber
Radio Number	1206

Primary Employee: Ethan C Barber [Make This Employee Primary](#) [Create Temp Crew](#)

Current crew employees

ID	NAME	TEMPCREWNAME	RADIONUMBER	CREWAREA	DEFAULTCREWAREA	EMPLOYEEYTYPE
FI1555	Jim Chris Carino		1208	N Jupiter District Crews	N Jupiter District Crews	Local Lineman
FI1601	Jon C Joiner		1210	N Jupiter District Crews	N Jupiter District Crews	Local Lineman
FI1560	K Geoff Gowdy		1209	N Jupiter District Crews	N Jupiter District Crews	Local Lineman

Filter: [Split Crew](#)

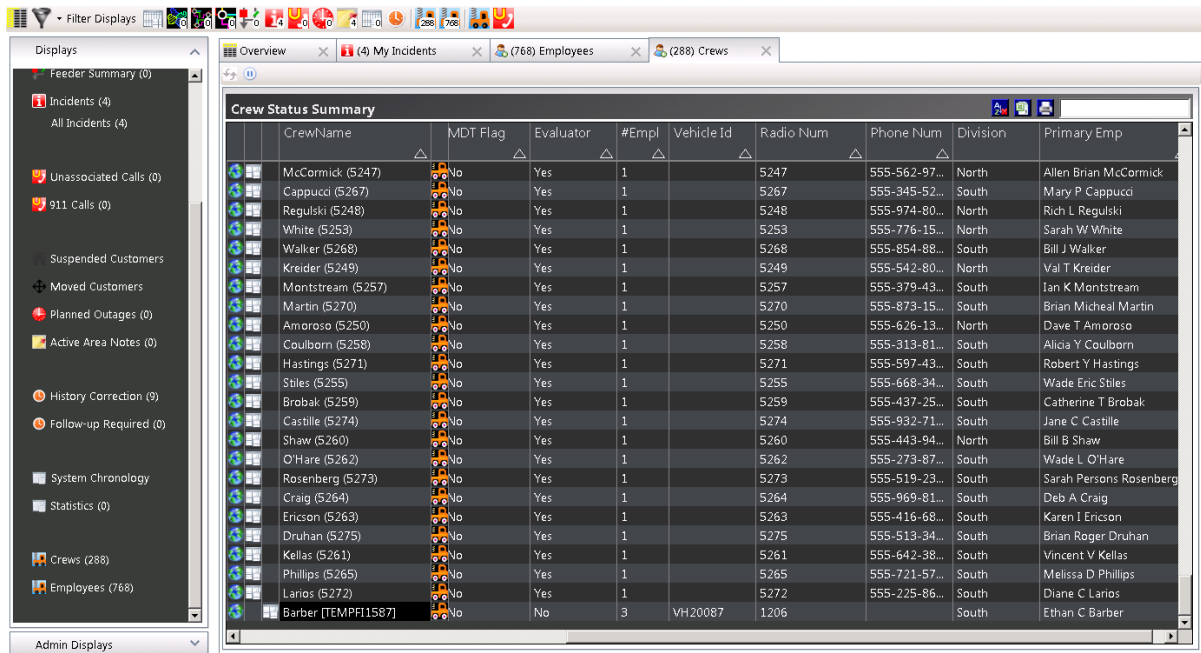
Available employees

ID	NAME	CREWNAME	TEMPCREWNAME	RADIONUMBER	CREWAREA	DEFAULTCREWAREA	EMPLOYEEYTYPE
FI2720	Kathy D Brobak				N Jupiter District Crews	N Jupiter District Crews	Manager
FI1500	Q Jeff Carino	Carino (1207)		1207	N Jupiter District Crews	N Jupiter District Crews	Local Lin

[Employees](#) [Vehicles](#) [OK](#) [Cancel](#) [Apply](#)

5. Click OK.

The temporary crew record is added at the bottom of the Crew Status Summary display.



For more information about the Crews tab, refer to the *Outage Management System User Guide*.

3.3. Managing Employees

You can add, delete, and maintain the crew and employee data using the Crew Modeling application.

3.3.1. Adding a New Employee

To add a new employee:

1. Open the Employees tab.
2. Right-click anywhere on the Employee Status grid and select the Add Employee option.

The Add Employee pane appears

3. Input the new employee's information.

Add Employee

Properties

Name	Value
Automatic Dispatch Id	
Capability	CanWorkOnIncident
Crew Area Id	
Crew Division	<None>
Crew Name	
Current Crew Area	<None>
Current Lead Name	
Default Crew Area	<None>
Default Crew Area Id	
Default Lead Name	
Default Radio Number	
Department	
Department Id	
Employee First Name	
Employee Last Name	
Employee Type	<None>
Employee Type Id	
Id	8288e87b-68aa-4c05-b8f2-2
Is Available	False

Cancel OK

4. Click OK.

The new employee record appears in the Employee Status grid.

3.3.2. Updating an Employee Record

To update an employee record:

1. Open the Employees tab.
2. Select the employee record you need to update.

The Update Employee pane appears populated with the selected record information.

Update Employee

Properties

Name	Value
Automatic Dispatch Id	11206
Capability	CanWorkOnIncident
Crew Area Id	LL1035
Crew Division	DispatchSouth
Crew Name	Ethan C Barber
Current Crew Area	N Jupiter District Crews
Default Crew Area	N Jupiter District Crews
Default Lead Name	<None>
Department	Dist Ops
Employee First Name	Ethan C
Employee Last Name	Barber
Employee Type	Local Lineman
Employee Type Id	3100
Id	FI1587
Is Available	True
Is Contractor	False

Cancel

OK

3. Update as needed and click OK to save the changes.

3.3.2.1. Moving Employees among Permanent Crews

The Crew Modeling tool creates a default crew for every employee whose employee type specifies that a crew should be created. This employee is the "primary employee" of the crew. An employee can be added to a default crew by setting the Lead field of an employee to an employee that has an automated crew.

If the employee's Lead field is set to <None>, the employee will not belong to any crew. If the employee's Lead field is set to a value selected from a combo box, the employee will become part of the lead's crew.

There are different methods to reassign the lead.

- Change the Lead field on the Update Employee pane as described in section [Updating an Employee Record](#).
- Assign an employee to a new crew on the Update Crew pane as described in section [Updating a](#)

Crew.

3.3.3. Deleting an Employee

To delete an employee:

1. Open the Employees tab.
2. Right-click the employee record on the Employee Status tab and select the Delete Employee option.
3. The employee information is deleted.

3.3.4. Creating a Temporary Crew

This option is available only from the OMS Display Plug-in.

To create a new temporary crew:

1. In the OMS Display Plug-in, on the Employees tab, right-click the desired employee record and select the Create Temp Crew option.

The Update Crew dialog box appears. The selected employee becomes the primary employee for the new temporary crew.

2. Filter employees as needed to search for the desired employees. Only employees from the primary employee crew area are available.
3. Add employees and vehicles to the new temporary crew as needed.

Crew Update Dialog - Crew Name: Jackson [TEMPFI1808]

Properties

Name	Value
Active	True
Automatic Dispatch Id	
Company Name	
Contractor	False
Crew Area	Central Crews
Crew Division	South
Crew Status	SignOn
Crew Type	Cable Splicer
Default Location X	0
Default Location Y	0
Department	
Evaluator	False
Foreign	False
MDT	False
Name	Jackson [TEMPFI1808]
Phone Number	
Primary Employee Id	FI1808
Primary Employee Name	Alvin Jeff Jackson

Primary Employee: Alvin Jeff Jackson [Make This Employee Primary](#) [Create Temp Crew](#)

Current crew employees

ID	NAME	TEMP CREW NAME	RADIONUMBER	CREW AREA	DEFAULT CREW AREA
FI2018	A Ben Druhan		1407	West Crews	West Crews
FI1918	A Rusty Connell		4048	South Crews	South Crews
FI2718	Aaron Craig Cowley			Coast Pinehurst District Crews	Coast Pinehurst Dist

Filter: [Split Crew](#)

Available employees

ID	NAME	CREW NAME	TEMP CREW NAME	RADIONUMBER	CREW AREA
FI1950	A Paul Morgan	Morgan [5279] - AUTO		4264	Central Crews
FI2141	Aaron P Dorrell	Robinson [5255] - AUTO		4156	West Crews
FI1607	Aaron P Drost	Drost (1135)		1135	Neptune Distric
FI1857	Aaron Tad Murphy	Crawford [5251] - AUTO		4085	West Crews
FI1841	Aaron X Cappucci	Thompson [5274] - AUTO		4104	Northeast Crev
FI1529	Adam D Johnson	Johnson (1132)		1132	Muirfield Distri
FI2043	Adam N Simons	Simons [4452] - AUTO		1452	Central Crews
FI1947	Adrian V Brannon	Brannon [4345]		4105	Northeast Crev

[Employees](#) [Vehicles](#)

[OK](#) [Cancel](#) [Apply](#)

4. Click Apply.

The Make This Employee Primary button is now enabled. To change the primary employee, select the employee from the Current Crew Employees box and click this button.

5. Click OK. The new temporary crew is added.

For more information about the Crews tab, refer to the *Outage Management System User Guide*.

3.4. Command-Line Interface

The Crew Modeling application can be run from command line. The tool's command-line interface supports the following arguments.

/BPATH

Used to specify the path to the bitmap directory. This argument is required when running the tool from the command line.

/CODEDTRANS [PATH TO FILE]

Used to specify the path to a coded translation configuration file. This argument is required when running the tool from the command line.

/IMPORTXML [PATH TO FILE]

Imports crew data from an XML file.

/SAVE [PATH TO FILE]

Saves the imported crew data as a CREW file. For example:

```
> CrewModelingTool.exe /CODEDTRANS %IDMS_ROOT%\config\shared\CodedTranslationConfig.xml /BPATH
%IDMS_ROOT%\bitmaps /IMPORTXML %IDMS_ROOT%\models\crew\CrewFile.XML /SAVE %IDMS_ROOT%\models\crew
\CrewFile.Crew
```

/OVERWRITEIFEXISTS

Allows saving a crew file with the name of an already existing file.

The argument used to compile only landbase information modified since the last compilation.

Note

The -ModifiedOnly argument functionality has not been implemented yet.

/LOG [PATH TO FILE]

Writes all log messages to the specified text file.

/SILENT

If this parameter is used, the Crew Modeling tool is working in silent mode (without user interface). For example:

```
> CrewModelingTool.exe /CODEDTRANS %IDMS_ROOT%\config\shared\CodedTranslationConfig.xml /BPATH  
%IDMS_ROOT%\bitmaps /LOG %IDMS_ROOT%\Log\CrewToolLog.txt /SILENT
```

```
/ENCRYPTION_KEY_SCOPE <EncryptionKeyScope> /ENCRYPTION_KEY_VERSION <EncryptionKeyScope>  
/ENCRYPTION_KEY_STORAGE_TYPE <EncryptionKeyStorageType>
```

If these parameters are used, the Crew Modeling tool is creating encrypted crew model files. For example:

```
> CrewModelingTool.exe /CODEDTRANS %IDMS_ROOT%\config\shared\CodedTranslationConfig.xml /BPATH  
%IDMS_ROOT%\bitmaps /IMPORTXML %IDMS_ROOT%\models\crew\CrewFile.XML /SAVE %IDMS_ROOT%\models\crew  
\CrewFile_Encrypted.Crew /SILENT /OVERWRITEIFEXIST /ENCRYPTION_KEY_SCOPE KADMS /ENCRYPTION_KEY_VERSION  
1 /ENCRYPTION_KEY_STORAGE_TYPE DEFAULT
```

Note

The crew model encryption requires an encryption key to be defined in the system. For more information, refer to the *ADMS Installation Guide*.

4. Building the Initial Crew Model

This chapter details how to create the initial crew model from extracts of employee information available in a third-party system.

4.1. Building a Crew Model in Excel

This section describes how to create text files with crew information in Excel, and then import the files in the Crew Modeling application.

4.1.1. Preparing Import Files

You can quickly create a new crew model from information stored in an Excel file.

The following sections detail the data required to create a valid crew model.



Tip

For a list of column names needed to properly import files into the Crew Modeling tool, select the Export > Export TSV option from the File menu. Then open the exported text files with Excel.

4.1.1.1. Employee Types

The Employee Types file contains basic information about available employee types. Once defined, this information is modified only if a new job type is introduced. The example below shows the employee types contained in an Excel file.

	A	B	C	D	E	F	G
1	Id	Name	AutoGenerateCrew	DoNotImplement	IsEvaluator		
2	1100	Administrator	FALSE	FALSE	FALSE		
3	1105	Administrator, Sr.	FALSE	FALSE	FALSE		
4	1110	Business Office Assistant	FALSE	FALSE	FALSE		
5	1300	Business Office Specialist	FALSE	FALSE	FALSE		
6	1501	Business Office Manager	FALSE	FALSE	FALSE		
7	2001	Utility Assistant	FALSE	FALSE	FALSE		
8	2005	Appr Cable Splicer	FALSE	FALSE	FALSE		
9	2010	Cable Splicer	FALSE	FALSE	FALSE		
10	2015	Lead Cable Splicer	TRUE	FALSE	FALSE		
11	2055	Apprentice Lineman	FALSE	FALSE	FALSE		
12	2060	Lineman	FALSE	FALSE	FALSE		
13	2065	Lead Lineman	TRUE	FALSE	FALSE		

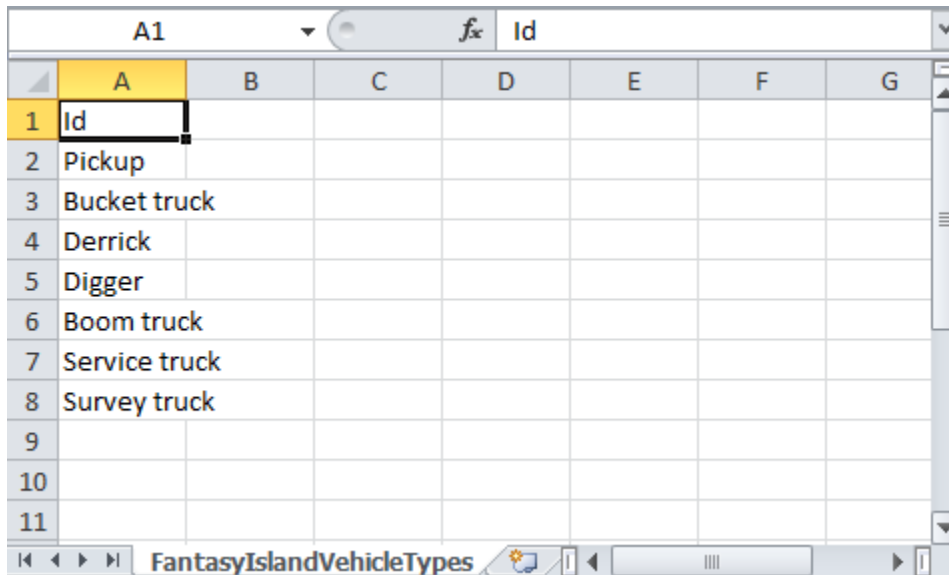
Figure 24. Employee Types in Excel

For descriptions of the column names, refer to section [Employee Types Tab](#).

4.1.1.2. Vehicle Types

The Vehicle Types file contains basic information about common vehicles the organization has. Once defined, this information is modified only if a new type of vehicle is introduced.

Currently there is only one column, "ID", which can be any string. In the example below, the ID is a short descriptive name.

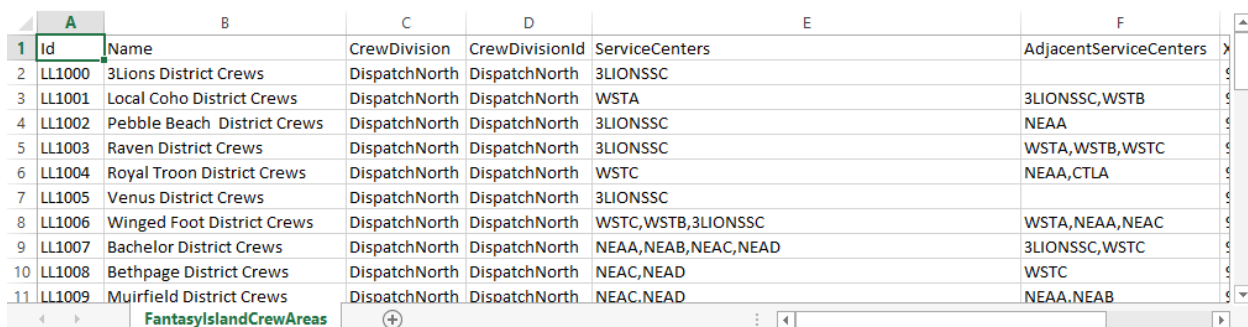


	A1						
	A	B	C	D	E	F	G
1	Id						
2	Pickup						
3	Bucket truck						
4	Derrick						
5	Digger						
6	Boom truck						
7	Service truck						
8	Survey truck						
9							
10							
11							

Figure 25. Example of Creating Vehicle Type IDs

4.1.1.3. Crew Areas

The Crew Areas file contains basic information about the organization's Crew Areas. Each crew normally works out of a specific work location, somewhere in the service area the crew normally works. Once defined, this information is modified only if a new crew area is added or an old crew area is discontinued.



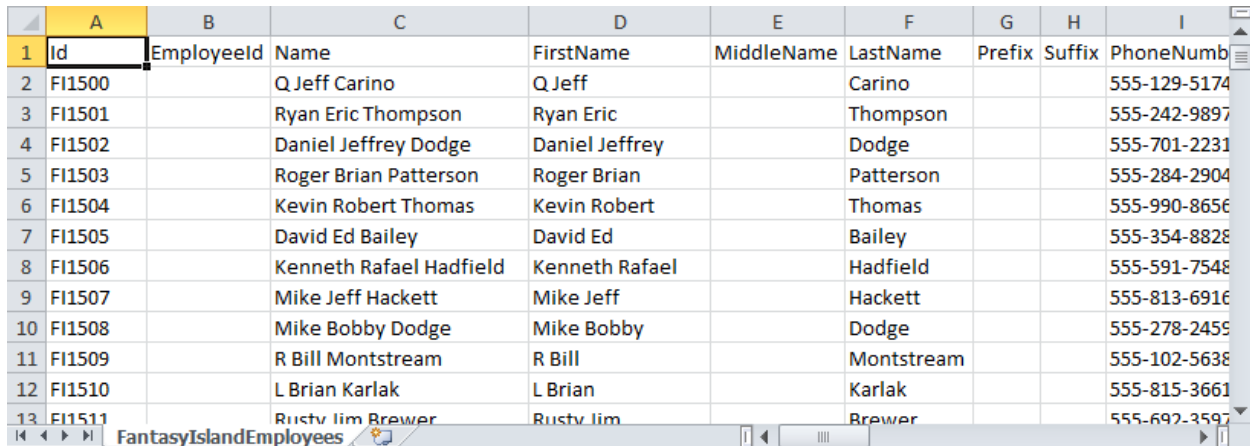
	A	B	C	D	E	F
	Id	Name	CrewDivision	CrewDivisionId	ServiceCenters	AdjacentServiceCenters
2	LL1000	3Lions District Crews	DispatchNorth	DispatchNorth	3LIONSSC	
3	LL1001	Local Coho District Crews	DispatchNorth	DispatchNorth	WSTA	3LIONSSC,WSTB
4	LL1002	Pebble Beach District Crews	DispatchNorth	DispatchNorth	3LIONSSC	NEAA
5	LL1003	Raven District Crews	DispatchNorth	DispatchNorth	3LIONSSC	WSTA,WSTB,WSTC
6	LL1004	Royal Troon District Crews	DispatchNorth	DispatchNorth	WSTC	NEAA,CTLA
7	LL1005	Venus District Crews	DispatchNorth	DispatchNorth	3LIONSSC	
8	LL1006	Winged Foot District Crews	DispatchNorth	DispatchNorth	WSTC,WSTB,3LIONSSC	WSTA,NEAA,NEAC
9	LL1007	Bachelor District Crews	DispatchNorth	DispatchNorth	NEAA,NEAB,NEAC,NEAD	3LIONSSC,WSTC
10	LL1008	Bethpage District Crews	DispatchNorth	DispatchNorth	NEAC,NEAD	WSTC
11	LL1009	Muirfield District Crews	DisoatchNorth	DisoatchNorth	NEAC,NEAD	NEAA,NEAB

Figure 26. Crew Areas in Excel

For descriptions of the column names, refer to section [Crew Areas Tab](#).

4.1.1.4. Employees

The Employees file contains the list of all employees that will be associated with a crew. This is the core information used to populate the crew model.



	A	B	C	D	E	F	G	H	I
1	Id	EmployeeId	Name	FirstName	MiddleName	LastName	Prefix	Suffix	PhoneNum
2	FI1500		Q Jeff Carino	Q Jeff		Carino			555-129-5174
3	FI1501		Ryan Eric Thompson	Ryan Eric		Thompson			555-242-9897
4	FI1502		Daniel Jeffrey Dodge	Daniel Jeffrey		Dodge			555-701-2231
5	FI1503		Roger Brian Patterson	Roger Brian		Patterson			555-284-2904
6	FI1504		Kevin Robert Thomas	Kevin Robert		Thomas			555-990-8656
7	FI1505		David Ed Bailey	David Ed		Bailey			555-354-8828
8	FI1506		Kenneth Rafael Hadfield	Kenneth Rafael		Hadfield			555-591-7548
9	FI1507		Mike Jeff Hackett	Mike Jeff		Hackett			555-813-6916
10	FI1508		Mike Bobby Dodge	Mike Bobby		Dodge			555-278-2459
11	FI1509		R Bill Montstream	R Bill		Montstream			555-102-5638
12	FI1510		L Brian Karlak	L Brian		Karlak			555-815-3661
13	FI1511		Rusty Jim Brewer	Rusty Jim		Brewer			555-697-3597

Figure 27. Example of Employees File

For descriptions of the column names, refer to section [Employees Tab](#).

The employee data is expected to be more dynamic than the "type" information described earlier. When a new employee is added to the organization, a new record is added with the employee's information. When the updated "Employee" file is imported into the Crew Modeling tool, the new employee is added to the crew model. If an employee is no longer working for the organization, the row pertaining to this employee should be removed from the Excel spreadsheet.

Note

Re-importing the "Employee" import file does not remove the employee from the crew model. The actual removal of the employee must be performed in the Crew Modeling tool using the Employees tab.

4.1.1.5. Vehicles

The Vehicles file contains the list of all vehicles to be associated with a crew. This is the core information used to populate the crew model's vehicle information.

	A	B	C	D	E	F	G	H	I	J	K	L
1	Id	Name	VehicleType	Id	Employee	Id	X	Y				
2	VH10000	Pickup 10000	Pickup			0	0					
3	VH10001	Pickup 10001	Pickup			0	0					
4	VH10002	Pickup 10002	Pickup			0	0					
5	VH10003	Pickup 10003	Pickup			0	0					
6	VH10004	Pickup 10004	Pickup			0	0					
7	VH10005	Pickup 10005	Pickup			0	0					
8	VH10006	Pickup 10006	Pickup			0	0					
9	VH10007	Pickup 10007	Pickup			0	0					
10	VH10008	Pickup 10008	Pickup			0	0					
11	VH10009	Pickup 10009	Pickup			0	0					
12	VH10010	Pickup 10010	Pickup			0	0					
13	VH10011	Pickup 10011	Pickup			0	0					

Figure 28. Example of Vehicles File

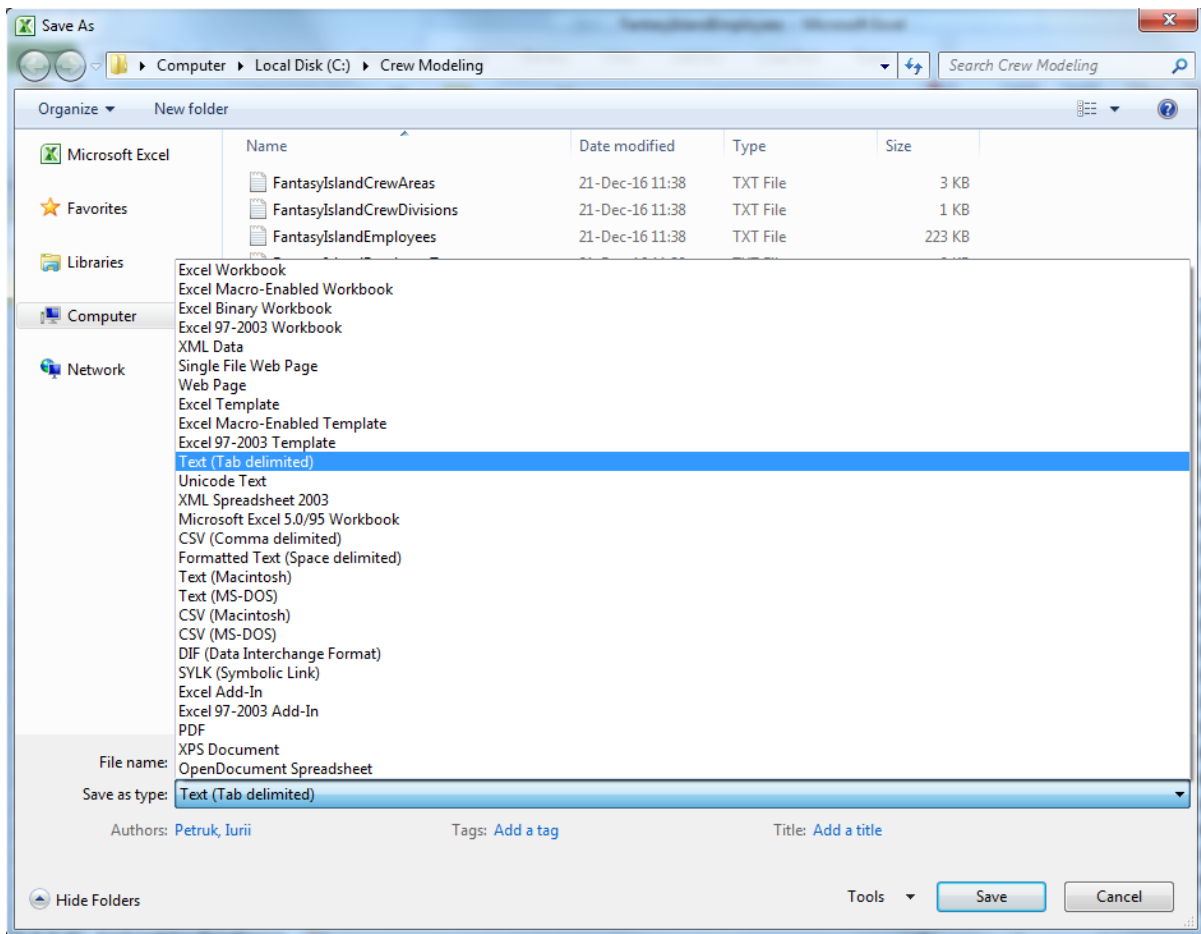
For descriptions of the column names, refer to section [Vehicles Tab](#).

The vehicle list is dynamic and needs to reflect the current state of the organization's vehicles. If a new vehicle is added, a new row can be added and re-imported into the Crew Modeling tool. If a vehicle is removed, it needs to be removed from the spreadsheet and also from the Crew Modeling tool's Vehicles tab.

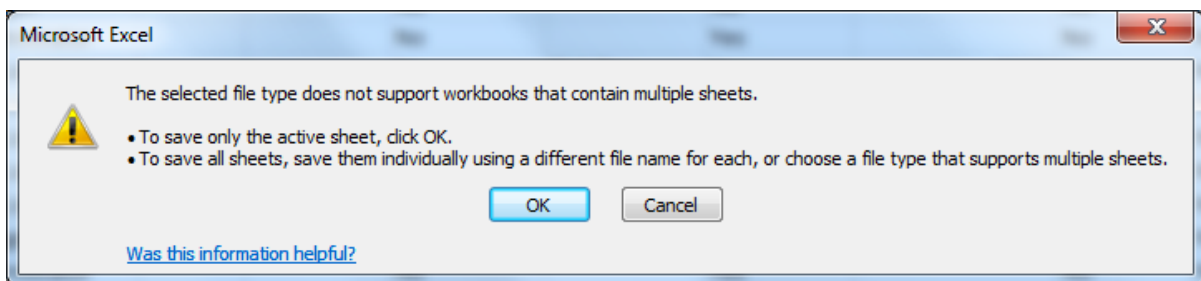
4.1.2. Creating Crew Modeling Import Files from Microsoft Excel

To convert an Excel file into a tab-delimited text file needed for importing the crew-related information into the offline Crew Modeling tool:

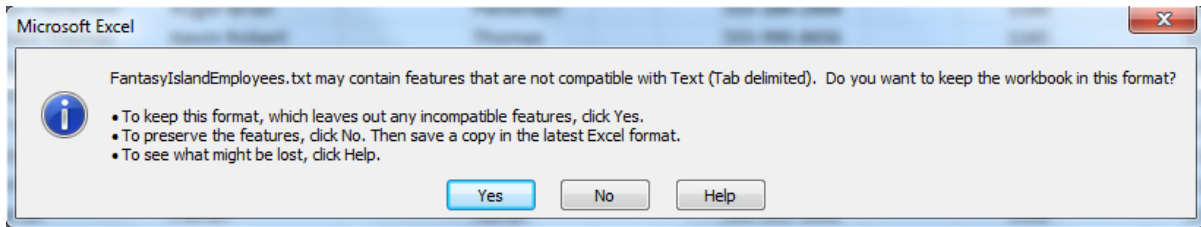
1. In Excel, select the tab containing the information you need to export. In the example below, the Employees information is selected.
2. From the File menu, select Save As.
3. In the Save As dialog box, select the type "Text (Tab delimited) (*.txt)".



4. Enter a file name and click Save.
5. At the dialog box with the message "The selected file type does not support workbooks that contain multiple sheets", click OK.



6. At the dialog box with the message "<filename>.txt may contain features that are not compatible with Text (Tab delimited). Do you want to keep the workbook in this format?", click Yes.



7. Repeat these steps to save files with the following information:

- Employee Types
- Vehicle Types
- Crew Areas
- Employees
- Vehicles

4.1.3. Importing Files into the Crew Modeling Tool

This section describes how to import files that were prepared in the previous section to create a new crew model.

The following files are ready to be imported:

- **EmployeeTypes.txt:** File containing information about the organization's employee types.
- **VehicleTypes.txt:** File containing information about the organization's vehicle types.
- **CrewAreas.txt:** File containing information about the organization's Crew Areas.
- **Vehicles.txt:** File containing information about each vehicle used by the organization's crews.
- **Employees.txt:** File containing information about each of the employees associated with crews.

Caution

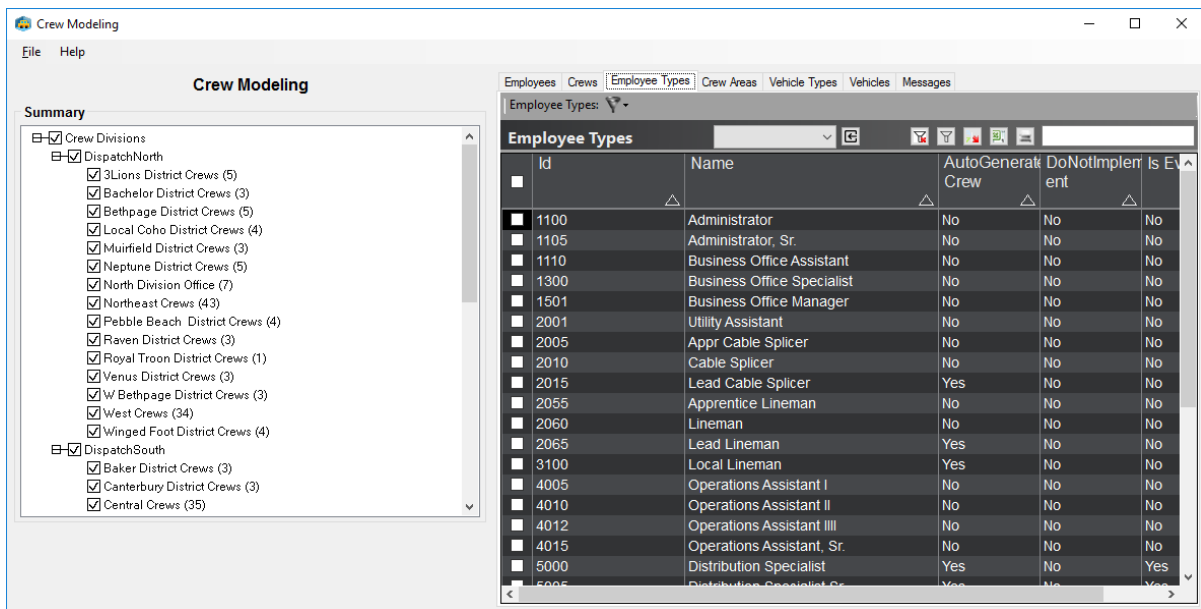
Improper data protection of the Employees.txt file may lead to exposing information that is not intended for disclosure. It is recommended that you configure authorization settings for this file so that only authorized applications and users can access it.

- **MutualAid.txt:** File containing information about foreign crew employees.

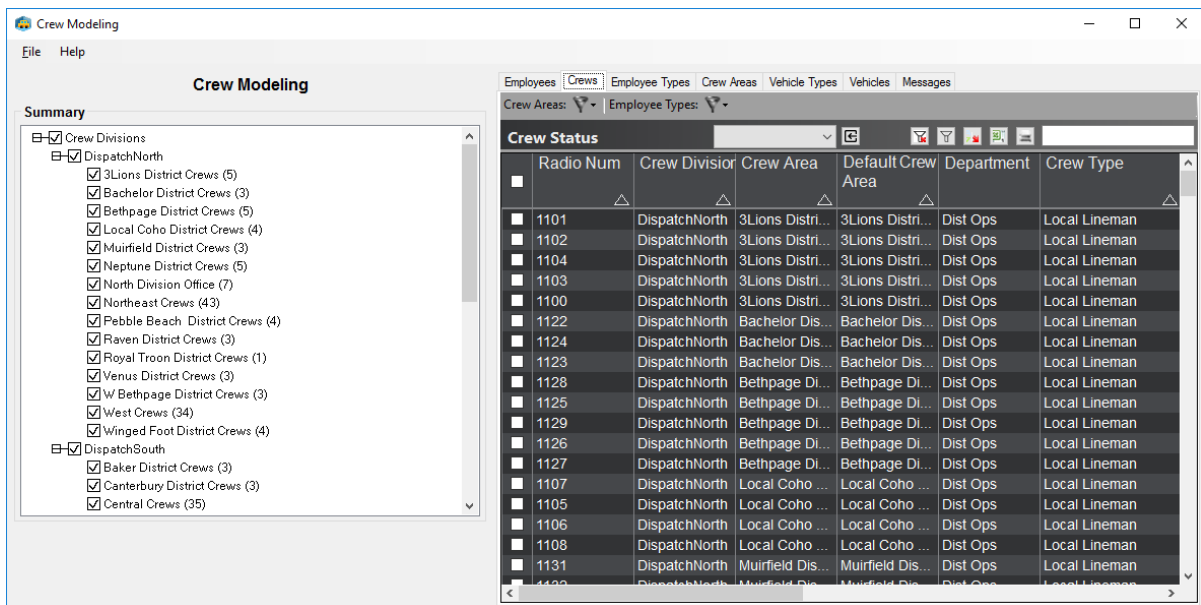
The files above are presented in the order they need to be imported into the Crew Modeling tool.

1. Start the Crew Modeling tool.
2. Select the Employee Type tab.
3. Click File > Import > Import Employee Types and open the EmployeeTypes file.

The Employee Types tab is populated with the file's data.



- Repeat the steps to select a corresponding tab and import each file in the order indicated above.
- Once the employees file is imported, crews are automatically created for all employees that are set as "primary employees" or with the Employee Type that has the AutoGenerateCrew flag set to True.



- Click File > Save As.
- In the Save As dialog box, enter the file name and location and click Save.

The new crew model is created.

4.2. Building a Crew Model in XML

This section describes how to create an XML file with crew information and then import the XML file in the Crew Modeling application.

4.2.1. Preparing an Import File

First, prepare an XML file with the crew information.



Tip

To obtain a sample XML import file, export the crew modeling data from the sample Fantasy Island crew model via the File > Export > XML menu command.

```
<?xml version="1.0" encoding="utf-8"?>
<EmployeeManagement xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="
"http://www.w3.org/2001/XMLSchema-instance">
  <Employees>
    <Employee>
      <Id>FI1500</Id>
      <EmployeeId />
      <FirstName>Q Jeff</FirstName>
      <MiddleName />
      <LastName>Carino</LastName>
      <Prefix />
      <Suffix />
      <PhoneNumber>555-129-5174</PhoneNumber>
      <SecondaryPhoneNumber />
      <RadioNumber>1207</RadioNumber>
      <AutomaticDispatchId>11207</AutomaticDispatchId>
      <DefaultCrewArea>N Jupiter District Crews</DefaultCrewArea>
      <CrewArea>N Jupiter District Crews</CrewArea>
      <CrewAreaId>LL1035</CrewAreaId>
      <SupervisorId />
      <SupervisorName />
      <DefaultLeadId />
      <CurrentLeadId />
      <EmployeeTypeId>3100</EmployeeTypeId>
      <EmployeeTypeName>Local Lineman</EmployeeTypeName>
    </Employee>
  </Employees>
</EmployeeManagement>
```

Figure 29. Example Crew Information

4.2.2. Validating an Import File

Use the following crew import schema to validate the crew import XML file.



Important

To enable integration with the Crew Server web interface and Workforce Management System,

set the MDTFlag field to True and populate the AutomaticDispatchId field for primary employees of crews that are equipped with a Mobile Data Terminal (MDT).

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema id="EmployeeManagement" xmlns="" xmlns:xs="http://www.w3.org/2001/XMLSchema"
xmlns:msdata="urn:schemas-microsoft-com:xml-msdata">
  <xs:annotation>
    <xs:appinfo source="urn:schemas-microsoft-com:xml-msdatasource">
      <DataSource DefaultConnectionIndex="0" FunctionsComponentName="QueriesTableAdapter"
Modifier="AutoLayout, AnsiClass, Class, Public" SchemaSerializationMode="IncludeSchema"
xmlns="urn:schemas-microsoft-com:xml-msdatasource">
        <Connections />
        <Tables />
        <Sources />
      </DataSource>
    </xs:appinfo>
  </xs:annotation>
  <xs:element name="EmployeeManagement" msdata:IsDataSet="true" msdata:UseCurrentLocale="true">
    <xs:complexType>
      <xs:choice minOccurs="0" maxOccurs="unbounded">
        <xs:element name="Employees">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="Employee" minOccurs="0" maxOccurs="unbounded">
                <xs:complexType>
                  <xs:sequence>
                    <xs:element name="Id" type="xs:string" />
                    <xs:element name="EmployeeId" type="xs:string" minOccurs="0" />
                    <xs:element name="FirstName" type="xs:string" />
                    <xs:element name="MiddleName" type="xs:string" minOccurs="0" />
                    <xs:element name="LastName" type="xs:string" />
                    <xs:element name="Prefix" type="xs:string" minOccurs="0" />
                    <xs:element name="Suffix" type="xs:string" minOccurs="0" />
                    <xs:element name="PhoneNumber" type="xs:string" />
                    <xs:element name="SecondaryPhoneNumber" type="xs:string"
minOccurs="0" />
                    <xs:element name="RadioNumber" type="xs:string" />
                    <xs:element name="AutomaticDispatchId" type="xs:string" />
                    <xs:element name="DefaultCrewArea" type="xs:string" />
                    <xs:element name="CrewArea" type="xs:string" />
                    <xs:element name="CrewAreaId" type="xs:string" />
                    <xs:element name="SupervisorId" type="xs:string" minOccurs="0" />
                    <xs:element name="SupervisorName" type="xs:string" minOccurs="0"
/>
                    <xs:element name="DefaultLeadId" type="xs:string" minOccurs="0" />
                    <xs:element name="CurrentLeadId" type="xs:string" minOccurs="0" />
                    <xs:element name="EmployeeTypeId" type="xs:string" />
                    <xs:element name="EmployeeTypeName" type="xs:string" />
                    <xs:element name="Dept" type="xs:string" minOccurs="0" />
                    <xs:element name="DeptId" type="xs:string" minOccurs="0" />
                    <xs:element name="VehicleTypeId" type="xs:string" minOccurs="0" />
                    <xs:element name="CrewDivision" type="xs:string" />

```

```

<xs:element name="IsEvaluator" type="xs:string" />
<xs:element name="IsExcluded" type="xs:string" minOccurs="0" />
<xs:element name="MembersCount" type="xs:string" minOccurs="0" />
<xs:element name="DivisionTypeCode" type="xs:string" minOccurs="0"
/>

<xs:element name="OpCenter" type="xs:string" minOccurs="0" />
<xs:element name="EmployeeTypeCode" type="xs:string" minOccurs="0"
/>

<xs:element name="ShoeSize" type="xs:string" minOccurs="0" />
<xs:element name="IsContractor" type="xs:string" />
<xs:element name="CapabilityCode" type="xs:string" minOccurs="0"
/>

<xs:element name="CanSwitch" type="xs:string" minOccurs="0" />
<xs:element name="EMPHQ" type="xs:string" minOccurs="0" />
<xs:element name="CrewName" type="xs:string" />
<xs:element name="CrewId" type="xs:string" />
<xs:element name="TempCrewName" type="xs:string" minOccurs="0" />
<xs:element name="IsPrimary" type="xs:string" />
<xs:element name="VehicleTypeCode" type="xs:string" minOccurs="0"
/>

<xs:element name="MDTFlag" type="xs:string" />
<xs:element name="IsForeign" type="xs:string" />
<xs:element name="UserId" type="xs:string" minOccurs="0" />
<xs:element name="Vehicles" minOccurs="0" maxOccurs="unbounded">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="Vehicle" minOccurs="0"
maxOccurs="unbounded">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="Id" type="xs:string" />
            <xs:element name="Name" type="xs:string"
minOccurs="0" />
            <xs:element name="VehicleTypeId"
type="xs:string" />
            <xs:element name="EmployeeId"
type="xs:string" minOccurs="0" />
            <xs:element minOccurs="0" maxOccurs="1"
name="CrewId" type="xs:string" />
            <xs:element minOccurs="0" maxOccurs="1"
name="PhoneNumber" type="xs:string" />
            <xs:element name="RadioNumber"
type="xs:string" minOccurs="0" />
            <xs:element minOccurs="0" />
            <xs:element name="X" type="xs:string"
minOccurs="0" />
            <xs:element name="Y" type="xs:string"
minOccurs="0" />
            <xs:element name="ReportedTime"
type="xs:string" minOccurs="0" />
            <xs:element name="UserId" type="xs:string"
minOccurs="0" />
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>

```

```

        </xs:element>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
</xs:sequence>
</xs:complexType>
</xs:element>
<xs:element name="CrewDivisions">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="CrewDivision" minOccurs="0" maxOccurs="unbounded">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="Id" type="xs:string" />
            <xs:element name="Name" type="xs:string" minOccurs="0" />
            <xs:element name="CrewAreas" minOccurs="0" maxOccurs="unbounded">
              <xs:complexType>
                <xs:sequence>
                  <xs:element name="CrewArea" minOccurs="0"
maxOccurs="unbounded">
                    <xs:complexType>
                      <xs:sequence>
                        <xs:element name="Id" type="xs:string" />
                        <xs:element name="Name" type="xs:string"
minOccurs="0" />
                        <xs:element name="CrewDivision"
type="xs:string" />
                        <xs:element name="CrewDivisionId"
type="xs:string" />
                        <xs:element name="ServiceCenters"
type="xs:string" />
                        <xs:element name="AdjacentServiceCenters"
type="xs:string" minOccurs="0" />
                        <xs:element name="X" type="xs:string"
minOccurs="0" />
                        <xs:element name="Y" type="xs:string"
minOccurs="0" />
                      </xs:sequence>
                    </xs:complexType>
                  </xs:element>
                </xs:sequence>
              </xs:complexType>
            </xs:element>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>
<xs:element name="EmployeeTypes">

```

```

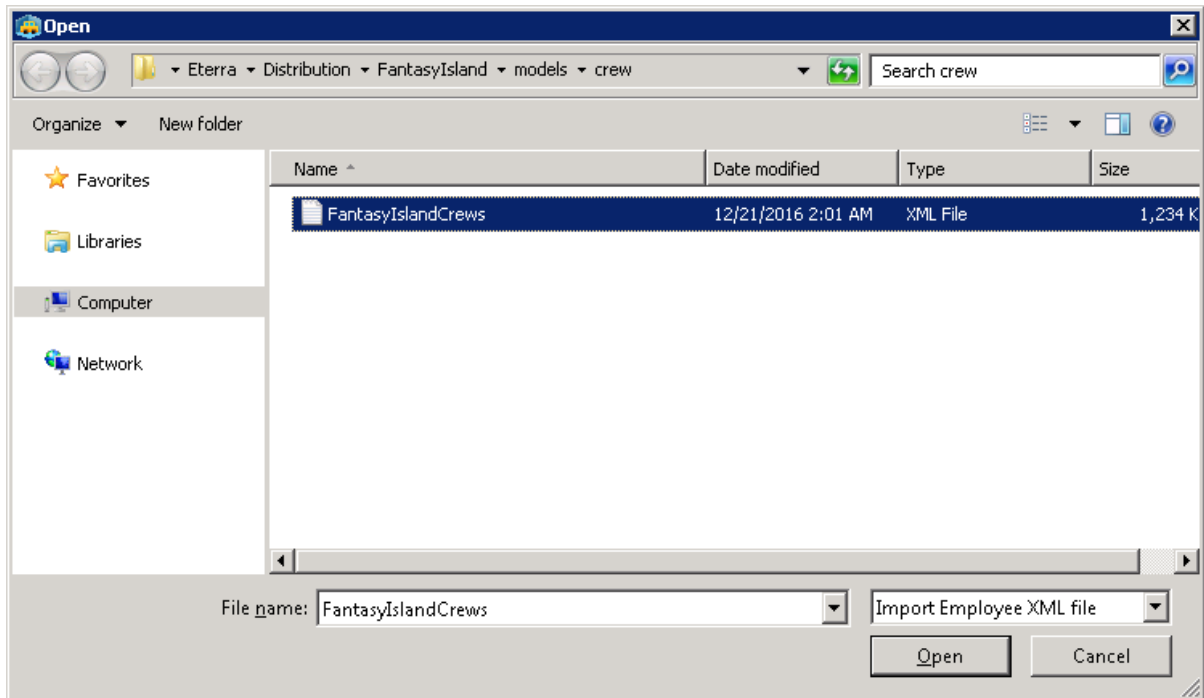
        <xs:complexType>
          <xs:sequence>
            <xs:element name="EmployeeType" minOccurs="0" maxOccurs="unbounded">
              <xs:complexType>
                <xs:sequence>
                  <xs:element name="Id" type="xs:string" />
                  <xs:element name="Name" type="xs:string" minOccurs="0" />
                  <xs:element name="AutoGenerateCrew" type="xs:string" />
                  <xs:element name="DoNotImplement" type="xs:string" minOccurs="0" />
                  <xs:element name="IsEvaluator" type="xs:string" minOccurs="0" />
                  <xs:element name="UserId" type="xs:string" minOccurs="0" />
                </xs:sequence>
              </xs:complexType>
            </xs:element>
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:choice>
  </xs:complexType>
</xs:element>
<xs:element name="VehicleTypes">
  <xs:complexType>
    <xs:sequence>
      <xs:element name="VehicleType" minOccurs="0" maxOccurs="unbounded">
        <xs:complexType>
          <xs:sequence>
            <xs:element name="Id" type="xs:string" minOccurs="0" />
            <xs:element name="UserId" type="xs:string" minOccurs="0" />
          </xs:sequence>
        </xs:complexType>
      </xs:element>
    </xs:sequence>
  </xs:complexType>
</xs:element>
</xs:choice>
</xs:complexType>
</xs:element>
</xs:schema>

```

4.2.3. Importing the Crew XML into the Crew Modeling Tool

To import a crew XML file to create a crew model:

1. Start the Crew Modeling tool.
2. Click File > Import > Import XML and open the XML file.



3. Click File > Save As.
4. In the Save As dialog box, enter the file name and location and click Save.

The new crew model is created.

5. Encrypting Crew Models

Note

The crew model encryption requires an encryption key to be defined in the system. For more information, refer to the *ADMS Installation Guide*.

The Crew Modeling tool can create encrypted crew model files.

To enable encryption:

1. Open the Crew Modeling Tool shortcut properties.
2. Add the following commands to the Target field of the shortcut:

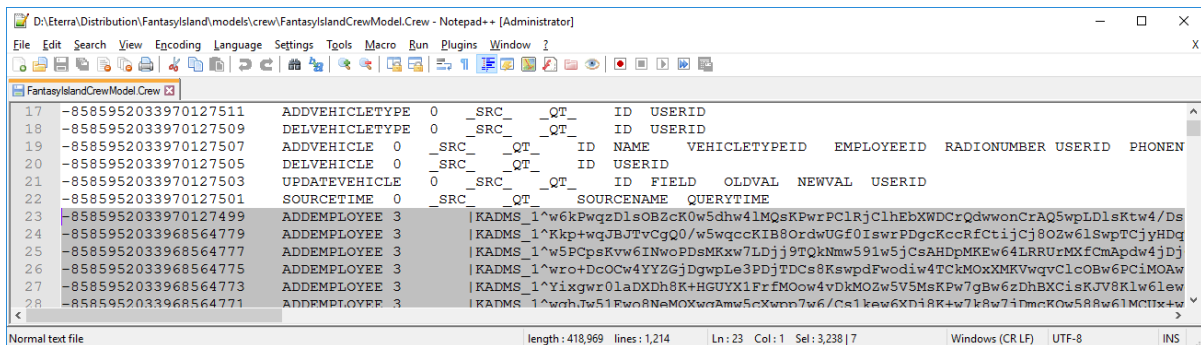
```
/ENCRYPTION_KEY_SCOPE <EncryptionKeyScope> /ENCRYPTION_KEY_VERSION <EncryptionKeyVersion>
```

For example:

```
D:\Eterra\Distribution\Tools\bin\CrewModelingTool.exe /CODEDTRANS  
"%IDMS_ROOT%\config\shared\CodedTranslationConfig.xml" /BPATH "%IDMS_ROOT%\bitmaps"  
/ENCRYPTION_KEY_SCOPE KADMS /ENCRYPTION_KEY_VERSION 1
```

3. Save the Crew Modeling Tool shortcut.

The Crew Modeling Tool will save encrypted crew model files. (The header information is not encrypted by design.)



Note

To use encrypted files, the same encryption key must be defined for users that encrypts files and for users that run applications which use the encrypted files.